

# Onyx — AppKit reference

Onyx · multi-process OS on Raspberry Pi 4

The reference of **AppKit** (`SD:/lib/appkit.so`, `user/Kits/appkit`): what it is for, how a program uses it, and every operation it exposes. The operations' part is made from the kit's headers by `tools/docgen/kitdocs.py` — the headers are the source. Overview of all the kits: [The Kits](#).

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## What it is

AppKit is what makes a program run: its one link to the system. Every call a program makes to the kernel is a function of AppKit (`kapi_*`), and AppKit carries the small services every program needs — strings, the console, `.ini` files, the keyboard layout, the starting of programs. Only AppKit reads the kernel's table, so the kernel can change without a program being rebuilt.

|         |                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------|
|         |                                                                                                                        |
| Include | <code>#include "appkit/appkit.h"</code>                                                                                |
| Link    | nothing to link: the kernel binds AppKit to every program                                                              |
| Library | <code>SD:/lib/appkit.so</code> — 305 entries in its table<br>( <code>user/Kits/appkit/appkit.abi</code> , append-only) |
| Sources | <code>user/Kits/appkit/</code>                                                                                         |

## Using it

AppKit is the program's link to the system: files, processes, time, windows, sockets... (the `kapi_*` calls, listed in the Developer Guide), plus the small services every program needs: strings without a C library, the console, a reader of `.ini` files, the starting of other programs.

**A console tool:** read a file, print to the console.

```
#include "appkit/appkit.h"
```

```
int main (void)
{
    char buf[256];
    void *f = kapi_open ("SD:/etc/autostart");
    if (f == 0) { ax_putln ("no such file"); return 1; }
    int n = kapi_read (f, buf, sizeof buf - 1);
    kapi_close (f);
    buf[n > 0 ? n : 0] = '\0';
}
```

```

ax_putln ("SD:/etc/autostart says:");
ax_putln (buf);
return 0;
}

```

**Settings from an .ini file** in the program's own folder:

```

if (app_ini_load ("config.ini") >= 0)
{
    const char *name = app_ini_get ("window", "title", "Untitled");
    int width      = app_ini_get_int ("window", "width", 640);
    for (int i = 0; i < app_ini_count (); i++)           // or every entry, in the
file's order
        ax_putln (app_ini_key (i));
}

```

**Strings** (no C library needed):

```

char path[96]; int n = 0;
ax_strcat (path, sizeof path, &n, "SD:/docs/");           // never overflows, always
terminated
ax_strcat (path, sizeof path, &n, "notes.txt");
char num[12]; ax_itoa (42, num);
if (ax_streq (num, "42")) ax_putln (path);

```

**Starting a program** — an application by its name, or a file by whatever runs it (a .bas program by BASIC, a game by its emulator):

```

lx_launch ("tinypad", "SD:/docs/notes.txt");           // an app, with its arguments
lx_open ("SD:/games/tetris.gb", 0);                     // a file: by its runner

```

**The volumes, USB sticks** (kapi v93) — list them, eject a stick, format one:

```

struct kapi_volume v[16];
int n = kapi_vol_list (v, 16, KAPI_VOLS_ROOM);           // SD:, SD1:..., USB1:...,
USB1P1:..., RAM:
for (int i = 0; i < n && i < 16; i++)
    if ((v[i].flags & KAPI_VF_REMOVABLE) && v[i].state == KAPI_VST_MOUNTED)
        ax_putln (v[i].label);                           // a stick plugged in: "USB" +
its label

if (kapi_vol_eject ("USB1:", 0) == -KAPI_EBUSY)           // files open on it (they were
synced)
    kapi_vol_eject ("USB1:", KAPI_EJECT_FORCE);           // ... after asking the user

struct kapi_format f = { KAPI_FMT_EXFAT, 0, 0, "PHOTOS" };
kapi_vol_format ("USB1:", &f);                           // erases it; SD: is always
refused

```

## Index

Everything the headers declare, in their order — the details are in each header's part below.

| Name             | What it does                                                                  | Header   |
|------------------|-------------------------------------------------------------------------------|----------|
| kapi_abi_version | The kernel's kapi version<br>(KAPI_ABI_VERSION as the running kernel says it) | appkit.h |

| Name                               | What it does                                                                                                                                                    | Header                |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>kapi_table_slot</code>       | The address of a slot of the kernel's table (its index in 8-byte words)                                                                                         | <code>appkit.h</code> |
| <code>kapi_create_window</code>    | This process's window (one a process), a client area of w x h pixels titled t, placed by the system → its canvas (0x00RRGGBB pixels), 0 on failure (bigger than | <code>appkit.h</code> |
| <code>kapi_create_window_ex</code> | This process's window at x, y (the frame's top left, negative = placed by the system) with the flags f (WIN_FLAG_*) → its canvas, 0 on failure.                 | <code>appkit.h</code> |
| <code>kapi_resize_window</code>    | This window's size set to w x h, within the canvas it was created with (which stays) → the canvas, 0 no window.                                                 | <code>appkit.h</code> |
| <code>kapi_move_window</code>      | this window's frame moved to x, y (screen coordinates)                                                                                                          | <code>appkit.h</code> |
| <code>kapi_launch</code>           | start the app n (SD:apps/.app/main) as a new process → 1, 0 failure                                                                                             | <code>appkit.h</code> |
| <code>kapi_toggle_app</code>       | Toggle the app n → 0 it was running and is asked to close (its window's exit flag), 1 it was started, -1 on error.                                              | <code>appkit.h</code> |
| <code>kapi_raise_app</code>        | The running app n's window (one on the current workspace) to the front → 1, 0 not running / no window.                                                          | <code>appkit.h</code> |
| <code>kapi_list_windows</code>     | The names of the open apps (a window on the current workspace, not a WIN_FLAG_SYSTEM one), one a line, into b (s bytes) → how many.                             | <code>appkit.h</code> |
| <code>kapi_list_tasks</code>       | Every task, one line each " " (state R / S / B / N, kind a = an app or k                                                                                        | <code>appkit.h</code> |

| Name                                 | What it does                                                                                                          | Header                |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------|
|                                      | = a kernel task), into b (s bytes) → how many.                                                                        |                       |
| <code>kapi_kill</code>               | kill the app of that name → 1, 0 (not running, a kernel task, the caller)                                             | <code>appkit.h</code> |
| <code>kapi_list_procs</code>         | ps / kill by PID.                                                                                                     | <code>appkit.h</code> |
| <code>kapi_kill_pid</code>           | ps / kill by PID.                                                                                                     | <code>appkit.h</code> |
| <code>kapi_proc_tree</code>          | (v91) A process's tree                                                                                                | <code>appkit.h</code> |
| <code>kapi_set_keymap</code>         | Keyboard layout                                                                                                       | <code>appkit.h</code> |
| <code>kapi_get_keymap</code>         | Keyboard layout                                                                                                       | <code>appkit.h</code> |
| <code>kapi_exec</code>               | Run an ELF at an absolute path with an argv string (fire-and-forget).                                                 | <code>appkit.h</code> |
| <code>kapi_screen_size</code>        | Framebuffer size in pixels (for edge-pinned borderless windows).                                                      | <code>appkit.h</code> |
| <code>kapi_wallpaper_generate</code> | Framebuffer size in pixels (for edge-pinned borderless windows).                                                      | <code>appkit.h</code> |
| <code>kapi_wallpaper_buffer</code>   | App-drawn wallpaper                                                                                                   | <code>appkit.h</code> |
| <code>kapi_wallpaper_commit</code>   | App-drawn wallpaper                                                                                                   | <code>appkit.h</code> |
| <code>kapi_present</code>            | App-drawn wallpaper                                                                                                   | <code>appkit.h</code> |
| <code>kapi_get_ticks</code>          | App-drawn wallpaper                                                                                                   | <code>appkit.h</code> |
| <code>kapi_msleep</code>             | App-drawn wallpaper                                                                                                   | <code>appkit.h</code> |
| <code>kapi_yield</code>              | App-drawn wallpaper                                                                                                   | <code>appkit.h</code> |
| <code>kapi_exit</code>               | App-drawn wallpaper                                                                                                   | <code>appkit.h</code> |
| <code>kapi_pump_events</code>        | Run what is pending -- the kapi_post calls, then this window's events through their handlers -- and return (no wait). | <code>appkit.h</code> |
| <code>kapi_wait_for_exit</code>      | pump the events (sleeping between them) until this window is asked to close                                           | <code>appkit.h</code> |
| <code>kapi_should_exit</code>        | 1 once this window was asked to close (its close box, kapi_toggle_app), else 0                                        | <code>appkit.h</code> |
| <code>kapi_draw_text</code>          | One line of text s in the kernel's font at x, y of this window's canvas, colour c (0x00RRGGBB), the                   | <code>appkit.h</code> |

| Name                                  | What it does                                                                                                          | Header                |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------|
|                                       | background kept.                                                                                                      |                       |
| <code>kapi_draw_text_buf</code>       | Draw kernel-font text into an arbitrary app-mapped 0x00RRGGBB buffer (e.g.                                            | <code>appkit.h</code> |
| <code>kapi_get_chrome</code>          | Window surfaces for a user-side chrome drawer (ABI v28).                                                              | <code>appkit.h</code> |
| <code>kapi_font_width</code>          | Window surfaces for a user-side chrome drawer (ABI v28).                                                              | <code>appkit.h</code> |
| <code>kapi_font_height</code>         | Window surfaces for a user-side chrome drawer (ABI v28).                                                              | <code>appkit.h</code> |
| <code>kapi_set_key_handler</code>     | Window surfaces for a user-side chrome drawer (ABI v28).                                                              | <code>appkit.h</code> |
| <code>kapi_set_click_handler</code>   | Window surfaces for a user-side chrome drawer (ABI v28).                                                              | <code>appkit.h</code> |
| <code>kapi_set_pointer_handler</code> | Window surfaces for a user-side chrome drawer (ABI v28).                                                              | <code>appkit.h</code> |
| <code>kapi_meminfo</code>             | Memory snapshot (KB)                                                                                                  | <code>appkit.h</code> |
| <code>kapi_ram_detail</code>          | ABI v33                                                                                                               | <code>appkit.h</code> |
| <code>kapi_set_wheel_speed</code>     | ABI v34                                                                                                               | <code>appkit.h</code> |
| <code>kapi_get_wheel_speed</code>     | ABI v34                                                                                                               | <code>appkit.h</code> |
| <code>kapi_sbrk</code>                | Per-process heap                                                                                                      | <code>appkit.h</code> |
| <code>kapi_list_apps</code>           | The names of the installed apps (the folders SD:apps/.app), one a line, into b (s bytes) → how many.                  | <code>appkit.h</code> |
| <code>kapi_get_datetime</code>        | The local date and time (any pointer may be 0) → 1 a real date, 0 the clock is not set yet (the time since the boot). | <code>appkit.h</code> |
| <code>kapi_app_dir</code>             | "SD:apps/<this process's name>.app/" into b (s bytes) → its length                                                    | <code>appkit.h</code> |
| <code>kapi_write</code>               | b (the first 128 bytes of its n) to the kernel's log as a line of "app" → n, -1 a bad buffer (fd is not               | <code>appkit.h</code> |

| Name                         | What it does                                                                                                                          | Header                |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|                              | used).                                                                                                                                |                       |
| <code>kapi_open</code>       | Open a file to read it (on the card, the RAM volume or a provider's path, relative to the working directory) → its handle, 0 failure. | <code>appkit.h</code> |
| <code>kapi_read</code>       | up to n bytes from the file's position (advanced) → the bytes read (0: the end), -1 error                                             | <code>appkit.h</code> |
| <code>kapi_fsize</code>      | the file's size in bytes (0xFFFFFFFF: over 4 GB, see <code>kapi_fsize64</code> ); 0 for a bad handle                                  | <code>appkit.h</code> |
| <code>kapi_close</code>      | close a file opened with <code>kapi_open</code>                                                                                       | <code>appkit.h</code> |
| <code>kapi_save_file</code>  | <code>save_file</code>                                                                                                                | <code>appkit.h</code> |
| <code>kapi_chdir</code>      | Working directory                                                                                                                     | <code>appkit.h</code> |
| <code>kapi_getcwd</code>     | Working directory                                                                                                                     | <code>appkit.h</code> |
| <code>kapi_opendir</code>    | open a folder to list it → its handle, 0 failure (not a folder)                                                                       | <code>appkit.h</code> |
| <code>kapi_readdir</code>    | the next entry (name, size, is_dir) into *e → 1, 0 at the end / on error                                                              | <code>appkit.h</code> |
| <code>kapi_closedir</code>   | close a folder opened with <code>kapi_opendir</code>                                                                                  | <code>appkit.h</code> |
| <code>kapi_mkdir</code>      | <code>mkdir</code> / <code>remove</code> / <code>rename</code>                                                                        | <code>appkit.h</code> |
| <code>kapi_remove</code>     | <code>mkdir</code> / <code>remove</code> / <code>rename</code>                                                                        | <code>appkit.h</code> |
| <code>kapi_rename</code>     | <code>mkdir</code> / <code>remove</code> / <code>rename</code>                                                                        | <code>appkit.h</code> |
| <code>kapi_cursor_pos</code> | <code>mkdir</code> / <code>remove</code> / <code>rename</code>                                                                        | <code>appkit.h</code> |
| <code>kapi_pipe</code>       | a new pipe (a FIFO in memory) → its stream handle, 0 failure                                                                          | <code>appkit.h</code> |
| <code>kapi_file_in</code>    | a file as a stream to read → its stream handle, 0 failure                                                                             | <code>appkit.h</code> |
| <code>kapi_file_out</code>   | A file as a stream to write, created / emptied or (append != 0) written                                                               | <code>appkit.h</code> |

| Name                             | What it does                                                                                                                                                                                  | Header                |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|                                  | at its end → its stream handle, 0 failure.                                                                                                                                                    |                       |
| <code>kapi_stream_read</code>    | Up to n bytes from a stream (a pipe waits for its writer) → the bytes read, 0 the end / a bad handle.                                                                                         | <code>appkit.h</code> |
| <code>kapi_stream_read_nb</code> | Up to n bytes from a stream without waiting → > 0 the bytes read, 0 the end / a bad handle, -1 nothing yet.                                                                                   | <code>appkit.h</code> |
| <code>kapi_stream_write</code>   | n bytes to a stream (a full pipe waits for its reader) → the bytes written, -1 error.                                                                                                         | <code>appkit.h</code> |
| <code>kapi_stream_close</code>   | close a stream handle (its reference to the stream dropped)                                                                                                                                   | <code>appkit.h</code> |
| <code>kapi_stream_eof</code>     | tell the stream's readers it has ended (the writer is done); the handle stays open                                                                                                            | <code>appkit.h</code> |
| <code>kapi_proc_done</code>      | 1 if the process of <code>kapi_spawn</code> has finished (a bad handle too), 0 if it runs (the handle stays, <code>kapi_wait</code> closes it).                                               | <code>appkit.h</code> |
| <code>kapi_stdin_read</code>     | up to n bytes from this process's standard input → the bytes read, 0 the end / none                                                                                                           | <code>appkit.h</code> |
| <code>kapi_stdout_write</code>   | n bytes to this process's standard output (it has none = to the kernel's log, 128 bytes at most) → the bytes written, -1 error.                                                               | <code>appkit.h</code> |
| <code>kapi_spawn</code>          | Start the program at path with the arguments args, its standard input / output the stream handles in / out (0 = none) → its process handle ( <code>kapi_wait</code> , <code>kapi_pro</code> ) | <code>appkit.h</code> |
| <code>kapi_wait</code>           | wait for a spawned                                                                                                                                                                            | <code>appkit.h</code> |

| Name                | What it does                                                                 | Header   |
|---------------------|------------------------------------------------------------------------------|----------|
|                     | process's end → its exit status (the handle closed), -1 not a process handle |          |
| kapi_get_args       | this process's argument string into b (n bytes) → its length                 | appkit.h |
| kapi_stdin          | This task's own stdin/stdout stream handles (for a shell wiring children).   | appkit.h |
| kapi_stdout         | This task's own stdin/stdout stream handles (for a shell wiring children).   | appkit.h |
| kapi_klog_read      | Read the next kernel log event (real-time tee).                              | appkit.h |
| kapi_set_verbose    | Verbose kernel logging                                                       | appkit.h |
| kapi_get_verbose    | Verbose kernel logging                                                       | appkit.h |
| kapi_net_status     | TCP/IP sockets over WLAN (ABI v21).                                          | appkit.h |
| kapi_tcp_connect    | TCP/IP sockets over WLAN (ABI v21).                                          | appkit.h |
| kapi_tcp_send       | TCP/IP sockets over WLAN (ABI v21).                                          | appkit.h |
| kapi_tcp_recv       | TCP/IP sockets over WLAN (ABI v21).                                          | appkit.h |
| kapi_tcp_close      | TCP/IP sockets over WLAN (ABI v21).                                          | appkit.h |
| kapi_tcp_listen     | TCP server side (ABI v37).                                                   | appkit.h |
| kapi_tcp_accept     | TCP server side (ABI v37).                                                   | appkit.h |
| kapi_screen_grab    | Remote screen (ABI v38).                                                     | appkit.h |
| kapi_inject_pointer | Remote screen (ABI v38).                                                     | appkit.h |
| kapi_inject_key     | Remote screen (ABI v38).                                                     | appkit.h |
| kapi_set_menu       | System menu bar (ABI v39).                                                   | appkit.h |
| kapi_get_menu       | System menu bar (ABI v39).                                                   | appkit.h |
| kapi_menu_command   | System menu bar (ABI v39).                                                   | appkit.h |

| Name                               | What it does                                                                                                                                                                                                       | Header                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>kapi_ipc_register</code>     | Named IPC services (ABI v40)                                                                                                                                                                                       | <code>appkit.h</code> |
| <code>kapi_ipc_lookup</code>       | Named IPC services (ABI v40)                                                                                                                                                                                       | <code>appkit.h</code> |
| <code>kapi_clipboard_set</code>    | The clipboard replaced by the <code>n</code> bytes of <code>d</code> (64 KB at most, <code>n 0</code> empties it) of the type <code>CLIP_*</code> → the bytes kept (0 too for a bad pointer, the clipboard then as | <code>appkit.h</code> |
| <code>kapi_clipboard_get</code>    | Up to <code>cap</code> bytes of the clipboard into <code>b</code> , its type and its serial (which changes at every set) → the content's whole length, 0 empty.                                                    | <code>appkit.h</code> |
| <code>kapi_set_window_alpha</code> | Window opacity 0..255 (ABI v40)                                                                                                                                                                                    | <code>appkit.h</code> |
| <code>kapi_shutdown</code>         | end the session: the card unmounted, then halt (SHUTDOWN_HALT) or restart; does not return                                                                                                                         | <code>appkit.h</code> |
| <code>kapi_fullscreen_begin</code> | Full-screen apps (ABI v41)                                                                                                                                                                                         | <code>appkit.h</code> |
| <code>kapi_present_fb</code>       | Full-screen apps (ABI v41)                                                                                                                                                                                         | <code>appkit.h</code> |
| <code>kapi_fullscreen_end</code>   | Full-screen apps (ABI v41)                                                                                                                                                                                         | <code>appkit.h</code> |
| <code>kapi_drag_begin</code>       | Drag & drop (ABI v42).                                                                                                                                                                                             | <code>appkit.h</code> |
| <code>kapi_drag_data</code>        | Drag & drop (ABI v42).                                                                                                                                                                                             | <code>appkit.h</code> |
| <code>kapi_get_modifiers</code>    | Drag & drop (ABI v42).                                                                                                                                                                                             | <code>appkit.h</code> |
| <code>kapi_inject_modifiers</code> | Drag & drop (ABI v42).                                                                                                                                                                                             | <code>appkit.h</code> |
| <code>kapi_net_ping</code>         | Network tools (ABI v43).                                                                                                                                                                                           | <code>appkit.h</code> |
| <code>kapi_net_resolve</code>      | Network tools (ABI v43).                                                                                                                                                                                           | <code>appkit.h</code> |
| <code>kapi_net_info</code>         | Network tools (ABI v43).                                                                                                                                                                                           | <code>appkit.h</code> |
| <code>kapi_vfs_register</code>     | this process serves the paths starting with prefix → 1, 0 (taken by another, no room)                                                                                                                              | <code>appkit.h</code> |
| <code>kapi_vfs_next</code>         | The next request for this provider into <code>*req</code>                                                                                                                                                          | <code>appkit.h</code> |

| Name                  | What it does                                                                                                                 | Header   |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------|----------|
|                       | → 1, 0 none (with blocking != 0, after waiting up to 0.5 s for one).                                                         |          |
| kapi_vfs_req_data     | Up to cap bytes of the request id's payload (VFS_OP_SAVE's data) from offset into buf → the bytes copied, 0 none.            | appkit.h |
| kapi_vfs_reply        | Answer the request id with its status and len bytes of data (0 = none), its caller woken → 1, 0 (no such request, bad data). | appkit.h |
| kapi_wlan_scan        | Wi-Fi scan (ABI v45)                                                                                                         | appkit.h |
| kapi_sound_acquire    | Sound (ABI v46), on the 3.5 mm jack.                                                                                         | appkit.h |
| kapi_sound_release    | Sound (ABI v46), on the 3.5 mm jack.                                                                                         | appkit.h |
| kapi_sound_start      | (RETIRED 2026-10-05)                                                                                                         | appkit.h |
| kapi_sound_stop       | (RETIRED 2026-10-05)                                                                                                         | appkit.h |
| kapi_sound_write      | (RETIRED 2026-10-05)                                                                                                         | appkit.h |
| kapi_sound_status     | (RETIRED 2026-10-05)                                                                                                         | appkit.h |
| kapi_sound_instrument | FM (ABI v47)                                                                                                                 | appkit.h |
| kapi_key_held         | Held keys (ABI v48), for games (key events only report presses)                                                              | appkit.h |
| kapi_inject_key_held  | Held keys (ABI v48), for games (key events only report presses)                                                              | appkit.h |
| kapi_exec_as          | Run a program under another process name (ABI v49)                                                                           | appkit.h |
| kapi_pad_state        | USB gamepads (v50)                                                                                                           | appkit.h |
| kapi_core_acquire     | App cores (v51)                                                                                                              | appkit.h |
| kapi_core_run         | App cores (v51)                                                                                                              | appkit.h |
| kapi_core_state       | App cores (v51)                                                                                                              | appkit.h |
| kapi_core_release     | App cores (v51)                                                                                                              | appkit.h |
| kapi_gpu_info         | The V3D GPU (v52)                                                                                                            | appkit.h |
| kapi_gpu_draw         | The V3D GPU (v52)                                                                                                            | appkit.h |
| kapi_gpu_texture      | The GPU's full pipeline (v53)                                                                                                | appkit.h |

| Name                                | What it does                                                                                                                                                                  | Header                |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>kapi_gpu_render</code>        | The GPU's full pipeline (v53)                                                                                                                                                 | <code>appkit.h</code> |
| <code>kapi_fullscreen_direct</code> | Full screen straight into the displayed framebuffer (v55)                                                                                                                     | <code>appkit.h</code> |
| <code>kapi_win_list</code>          | The windows as objects (v56, the remote desktop rdpd)                                                                                                                         | <code>appkit.h</code> |
| <code>kapi_win_read</code>          | The windows as objects (v56, the remote desktop rdpd)                                                                                                                         | <code>appkit.h</code> |
| <code>kapi_win_raise</code>         | The windows as objects (v56, the remote desktop rdpd)                                                                                                                         | <code>appkit.h</code> |
| <code>kapi_win_close</code>         | The windows as objects (v56, the remote desktop rdpd)                                                                                                                         | <code>appkit.h</code> |
| <code>kapi_seek</code>              | The read position of an opened file (v57)                                                                                                                                     | <code>appkit.h</code> |
| <code>kapi_code_alloc</code>        | Writable + executable memory for generated code, a JIT (v58)                                                                                                                  | <code>appkit.h</code> |
| <code>kapi_fsize64</code>           | A file's whole size (v59)                                                                                                                                                     | <code>appkit.h</code> |
| <code>kapi_vol_info</code>          | (v71) <code>vol_info</code>                                                                                                                                                   | <code>appkit.h</code> |
| <code>kapi_vol_list</code>          | (v93) The volumes                                                                                                                                                             | <code>appkit.h</code> |
| <code>kapi_vol_eject</code>         | <code>vol_eject</code>                                                                                                                                                        | <code>appkit.h</code> |
| <code>kapi_vol_mount</code>         | <code>vol_mount</code>                                                                                                                                                        | <code>appkit.h</code> |
| <code>kapi_vol_format</code>        | <code>vol_format</code>                                                                                                                                                       | <code>appkit.h</code> |
| <code>kapi_pop_event</code>         | (v73) The event pump's kernel half -- what <code>kapi_pump_events</code> does, step by step, for a pump of the app's own (a protected app's table runs its pump that way, ker | <code>appkit.h</code> |
| <code>kapi_event_mods</code>        | (v73) The event pump's kernel half -- what <code>kapi_pump_events</code> does, step by step, for a pump of the app's own (a protected app's table runs its pump that way, ker | <code>appkit.h</code> |
| <code>kapi_pop_post</code>          | (v73) The event pump's kernel half -- what                                                                                                                                    | <code>appkit.h</code> |

| Name                  | What it does                                                                                                                                                     | Header   |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                       | kapi_pump_events does, step by step, for a pump of the app's own (a protected app's table runs its pump that way, ker                                            |          |
| kapi_pump_sleep       | (v73) The event pump's kernel half -- what kapi_pump_events does, step by step, for a pump of the app's own (a protected app's table runs its pump that way, ker | appkit.h |
| kapi_proc_stats       | (v74) A process's system calls (pid 0                                                                                                                            | appkit.h |
| kapi_vm_map           | (v75) The POSIX layer's kernel half<br>(docs/POSIX-PLAN.md                                                                                                       | appkit.h |
| kapi_vm_unmap         | (v75) The POSIX layer's kernel half<br>(docs/POSIX-PLAN.md                                                                                                       | appkit.h |
| kapi_vm_protect       | (v75) The POSIX layer's kernel half<br>(docs/POSIX-PLAN.md                                                                                                       | appkit.h |
| kapi_vm_advise        | (v75) The POSIX layer's kernel half<br>(docs/POSIX-PLAN.md                                                                                                       | appkit.h |
| kapi_vm_query         | (v75) The POSIX layer's kernel half<br>(docs/POSIX-PLAN.md                                                                                                       | appkit.h |
| kapi_vm_stats         | (v75) The POSIX layer's kernel half<br>(docs/POSIX-PLAN.md                                                                                                       | appkit.h |
| kapi_thread_create_ex | (v75) The POSIX layer's kernel half<br>(docs/POSIX-PLAN.md                                                                                                       | appkit.h |
| kapi_thread_info      | (v75) The POSIX layer's kernel half<br>(docs/POSIX-PLAN.md                                                                                                       | appkit.h |
| kapi_file_open        | v75 WP-FILE/PROC                                                                                                                                                 | appkit.h |
| kapi_file_read        | v75 WP-FILE/PROC                                                                                                                                                 | appkit.h |
| kapi_file_write       | v75 WP-FILE/PROC                                                                                                                                                 | appkit.h |
| kapi_file_seek        | v75 WP-FILE/PROC                                                                                                                                                 | appkit.h |
| kapi_file_truncate    | v75 WP-FILE/PROC                                                                                                                                                 | appkit.h |
| kapi_file_sync        | v75 WP-FILE/PROC                                                                                                                                                 | appkit.h |

| Name                 | What it does     | Header   |
|----------------------|------------------|----------|
| kapi_file_stat       | v75 WP-FILE/PROC | appkit.h |
| kapi_file_close      | v75 WP-FILE/PROC | appkit.h |
| kapi_path_stat       | v75 WP-FILE/PROC | appkit.h |
| kapi_path_unlink     | v75 WP-FILE/PROC | appkit.h |
| kapi_path_mkdir      | v75 WP-FILE/PROC | appkit.h |
| kapi_path_rename     | v75 WP-FILE/PROC | appkit.h |
| kapi_path_utime      | v75 WP-FILE/PROC | appkit.h |
| kapi_dir_read        | v75 WP-FILE/PROC | appkit.h |
| kapi_stream_write_nb | v75 WP-FILE/PROC | appkit.h |
| kapi_spawn_ex        | v75 WP-FILE/PROC | appkit.h |
| kapi_proc_wait       | v75 WP-FILE/PROC | appkit.h |
| kapi_get_argv        | v75 WP-FILE/PROC | appkit.h |
| kapi_get_env         | v75 WP-FILE/PROC | appkit.h |
| kapi_getpid          | v75 WP-FILE/PROC | appkit.h |
| kapi_clock_info      | v75 WP-FILE/PROC | appkit.h |
| kapi_sleep_us        | v75 WP-FILE/PROC | appkit.h |
| kapi_sock_open       | v75 WP-NET       | appkit.h |
| kapi_sock_connect    | v75 WP-NET       | appkit.h |
| kapi_sock_bind       | v75 WP-NET       | appkit.h |
| kapi_sock_listen     | v75 WP-NET       | appkit.h |
| kapi_sock_accept     | v75 WP-NET       | appkit.h |
| kapi_sock_send       | v75 WP-NET       | appkit.h |
| kapi_sock_recv       | v75 WP-NET       | appkit.h |
| kapi_sock_shutdown   | v75 WP-NET       | appkit.h |
| kapi_sock_close      | v75 WP-NET       | appkit.h |
| kapi_sock_getopt     | v75 WP-NET       | appkit.h |
| kapi_sock_setopt     | v75 WP-NET       | appkit.h |
| kapi_sock_name       | v75 WP-NET       | appkit.h |
| kapi_poll            | v75 WP-NET       | appkit.h |
| kapi_sock_pair       | (v76, WP-IPC     | appkit.h |
| kapi_sock_sendmsg    | (v76, WP-IPC     | appkit.h |
| kapi_sock_recvmsg    | (v76, WP-IPC     | appkit.h |
| kapi_shm_create      | (v76, WP-IPC     | appkit.h |
| kapi_shm_open        | (v76, WP-IPC     | appkit.h |
| kapi_shm_unlink      | (v76, WP-IPC     | appkit.h |
| kapi_shm_ctl         | (v76, WP-IPC     | appkit.h |
| kapi_shm_map         | (v76, WP-IPC     | appkit.h |
| kapi_handle_close    | (v76, WP-IPC     | appkit.h |
| kapi_spawn_ex2       | (v76, WP-IPC     | appkit.h |

| Name                                  | What it does                                                                                                                         | Header                |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>kapi_get_handles</code>         | (v76, WP-IPC                                                                                                                         | <code>appkit.h</code> |
| <code>kapi_image_preload</code>       | (v77) Program images (docs/02 section 7)                                                                                             | <code>appkit.h</code> |
| <code>kapi_image_unload</code>        | (v77) Program images (docs/02 section 7)                                                                                             | <code>appkit.h</code> |
| <code>kapi_image_list</code>          | (v77) Program images (docs/02 section 7)                                                                                             | <code>appkit.h</code> |
| <code>kapi_kernel_info</code>         | (v79) What the running kernel is                                                                                                     | <code>appkit.h</code> |
| <code>kapi_cpu_stats</code>           | (v80) The cores                                                                                                                      | <code>appkit.h</code> |
| <code>kapi_net_stats</code>           | (v80) The bytes pid's sockets received and sent, its open sockets (pid 0                                                             | <code>appkit.h</code> |
| <code>kapi_set_cursor</code>          | (v81) The pointer's shape over this window (KAPI_CURSOR_*) → the shape it had                                                        | <code>appkit.h</code> |
| <code>kapi_win_resizable</code>       | this window resizable by its frame (on 0: no longer), min_w x min_h its smallest client area → 0, -1                                 | <code>appkit.h</code> |
| <code>kapi_lib_open</code>            | (v83) A shared library (docs/SHARED-LIBS-PLAN.md)                                                                                    | <code>appkit.h</code> |
| <code>kapi_sound_output</code>        | (v84) The sound's output                                                                                                             | <code>appkit.h</code> |
| <code>kapi_sound_clients</code>       | (v85) the sound's mixer                                                                                                              | <code>appkit.h</code> |
| <code>kapi_sound_client_volume</code> | (v85) the sound's mixer                                                                                                              | <code>appkit.h</code> |
| <code>kapi_is_protected</code>        | (v73) Is this process protected (ELO, kern/el0.h)? Its table's memcpy is then user code, next to the table, instead of the kernel's. | <code>appkit.h</code> |
| <code>kapi_sound_volume</code>        | the master volume 0..10 and mute (-1                                                                                                 | <code>appkit.h</code> |
| <code>kapi_wlan_reconnect</code>      | wpa_supplicant.conf read again + DHCP again, no reboot (-1                                                                           | <code>appkit.h</code> |
| <code>kapi_gpu_program</code>         | v61                                                                                                                                  | <code>appkit.h</code> |
| <code>kapi_gpu_render2</code>         | v61                                                                                                                                  | <code>appkit.h</code> |
| <code>kapi_gpu_render3</code>         | (v62) gpu_render3                                                                                                                    | <code>appkit.h</code> |

| Name                               | What it does                                                                                                                        | Header                |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>kapi_gpu_vbuf</code>         | (v63) <code>gpu_vbuf</code>                                                                                                         | <code>appkit.h</code> |
| <code>kapi_gpu_texture_rect</code> | (v70) <code>gpu_texture_rect</code>                                                                                                 | <code>appkit.h</code> |
| <code>kapi_win_minimise</code>     | (v64) the windows of the modernised CDE desktop                                                                                     | <code>appkit.h</code> |
| <code>kapi_win_geometry</code>     | (v64) the windows of the modernised CDE desktop                                                                                     | <code>appkit.h</code> |
| <code>kapi_resize_window2</code>   | (v64) the windows of the modernised CDE desktop                                                                                     | <code>appkit.h</code> |
| <code>kapi_desk</code>             | show the desk <code>set</code> (-1: keep) and set their number (0: keep) → <code>KAPI_DESK_CUR / _COUNT / _GEN</code> of the result | <code>appkit.h</code> |
| <code>kapi_win_desk</code>         | the window id (0: mine) moved to the desk <code>n</code> (-1: every desk; -2: only ask) → its desk, -3 no such window               | <code>appkit.h</code> |
| <code>kapi_screen_set</code>       | (v66) <code>screen_set</code>                                                                                                       | <code>appkit.h</code> |
| <code>kapi_thread_create</code>    | (v67) Threads                                                                                                                       | <code>appkit.h</code> |
| <code>kapi_thread_exit</code>      | (v67) Threads                                                                                                                       | <code>appkit.h</code> |
| <code>kapi_thread_join</code>      | (v67) Threads                                                                                                                       | <code>appkit.h</code> |
| <code>kapi_thread_self</code>      | (v67) Threads                                                                                                                       | <code>appkit.h</code> |
| <code>kapi_mutex_create</code>     | Synchronisation objects (handles > 0)                                                                                               | <code>appkit.h</code> |
| <code>kapi_mutex_lock</code>       | Synchronisation objects (handles > 0)                                                                                               | <code>appkit.h</code> |
| <code>kapi_mutex_unlock</code>     | Synchronisation objects (handles > 0)                                                                                               | <code>appkit.h</code> |
| <code>kapi_event_create</code>     | Synchronisation objects (handles > 0)                                                                                               | <code>appkit.h</code> |
| <code>kapi_event_set</code>        | Synchronisation objects (handles > 0)                                                                                               | <code>appkit.h</code> |
| <code>kapi_event_reset</code>      | Synchronisation objects (handles > 0)                                                                                               | <code>appkit.h</code> |
| <code>kapi_event_wait</code>       | Synchronisation objects (handles > 0)                                                                                               | <code>appkit.h</code> |
| <code>kapi_barrier_create</code>   | Synchronisation objects (handles > 0)                                                                                               | <code>appkit.h</code> |
| <code>kapi_barrier_wait</code>     | Synchronisation objects (handles > 0)                                                                                               | <code>appkit.h</code> |

| Name                               | What it does                                                                                                      | Header                |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>kapi_sync_close</code>       | Synchronisation objects (handles > 0)                                                                             | <code>appkit.h</code> |
| <code>kapi_post</code>             | Synchronisation objects (handles > 0)                                                                             | <code>appkit.h</code> |
| <code>kapi_pump_wait</code>        | Synchronisation objects (handles > 0)                                                                             | <code>appkit.h</code> |
| <code>kapi_lock</code>             | (on an app core -- <code>kapi_core_run</code> 's code, which makes no <code>kapi</code> call -- it spins instead) | <code>appkit.h</code> |
| <code>kapi_unlock</code>           | (on an app core -- <code>kapi_core_run</code> 's code, which makes no <code>kapi</code> call -- it spins instead) | <code>appkit.h</code> |
| <code>kapi_wait_word</code>        | (v68) A futex.                                                                                                    | <code>appkit.h</code> |
| <code>kapi_wake_word</code>        | (v68) A futex.                                                                                                    | <code>appkit.h</code> |
| <code>kapi_thread_priority</code>  | (v68) A futex.                                                                                                    | <code>appkit.h</code> |
| <code>kapi_midi_read</code>        | (v68) USB MIDI input                                                                                              | <code>appkit.h</code> |
| <code>kapi_midi_devices</code>     | (v68) USB MIDI input                                                                                              | <code>appkit.h</code> |
| <code>kapi_screen_native</code>    | (v69) <code>kapi_screen_native (&amp;w, &amp;h)</code>                                                            | <code>appkit.h</code> |
| <code>kapi_set_timezone</code>     | (v69) <code>kapi_screen_native (&amp;w, &amp;h)</code>                                                            | <code>appkit.h</code> |
| <code>kapi_clock_us</code>         | The kernel's microsecond clock ( <code>CTimer::GetClockTicks</code> )                                             | <code>appkit.h</code> |
| <code>kapi_sound_config</code>     | (v68) Low-latency sound, for the sound owner ( <code>kapi_sound_acquire</code> ).                                 | <code>appkit.h</code> |
| <code>kapi_sound_map</code>        | (v68) Low-latency sound, for the sound owner ( <code>kapi_sound_acquire</code> ).                                 | <code>appkit.h</code> |
| <code>kapi_sound_ring_free</code>  | Frames the ring can take now.                                                                                     | <code>appkit.h</code> |
| <code>kapi_sound_ring_write</code> | Copy up to <code>n</code> s16 stereo frames into the ring → the frames taken (no <code>kapi</code> call)          | <code>appkit.h</code> |
| <code>kapi_reboot</code>           | Reboot the machine (ABI v25).                                                                                     | <code>appkit.h</code> |
| <code>kapi_kbd_ready</code>        | 1 if a USB keyboard is                                                                                            | <code>appkit.h</code> |

| Name                              | What it does                                                                    | Header                |
|-----------------------------------|---------------------------------------------------------------------------------|-----------------------|
|                                   | attached & ready, else 0 (ABI v26).                                             |                       |
| <code>kapi_set_keymap_data</code> | Load a keyboard layout from a .kmap blob (ABI v27)                              | <code>appkit.h</code> |
| <code>kapi_random</code>          | Hardware RNG (ABI v30)                                                          | <code>appkit.h</code> |
| <code>kapi_surface_create</code>  | Shell surfaces (ABI v35)                                                        | <code>appkit.h</code> |
| <code>kapi_surface_map</code>     | Shell surfaces (ABI v35)                                                        | <code>appkit.h</code> |
| <code>kapi_surface_size</code>    | Shell surfaces (ABI v35)                                                        | <code>appkit.h</code> |
| <code>kapi_surface_present</code> | Shell surfaces (ABI v35)                                                        | <code>appkit.h</code> |
| <code>kapi_surface_destroy</code> | Shell surfaces (ABI v35)                                                        | <code>appkit.h</code> |
| <code>kapi_register_shell</code>  | Activity-shell IPC (ABI v35)                                                    | <code>appkit.h</code> |
| <code>kapi_shell_request</code>   | Activity-shell IPC (ABI v35)                                                    | <code>appkit.h</code> |
| <code>kapi_mailbox_send</code>    | Activity-shell IPC (ABI v35)                                                    | <code>appkit.h</code> |
| <code>kapi_mailbox_recv</code>    | Activity-shell IPC (ABI v35)                                                    | <code>appkit.h</code> |
| <code>kapi_cursor_shown</code>    | (v89) The pointer's shape shown now, whatever window it is over (KAPI_CURSOR_*) | <code>appkit.h</code> |
| <code>kapi_ws_ctl</code>          | (v89) The graphics server's own door to the kernel (Elegant, SD:/bin/elegant)   | <code>appkit.h</code> |
| <code>kapi_memset</code>          | Memory primitives (ABI v36)                                                     | <code>appkit.h</code> |
| <code>kapi_memcpy</code>          | Memory primitives (ABI v36)                                                     | <code>appkit.h</code> |
| <code>kapi_memmove</code>         | Memory primitives (ABI v36)                                                     | <code>appkit.h</code> |
| <code>create_window</code>        | Friendly aliases used by the demos.                                             | <code>appkit.h</code> |
| <code>present</code>              | Friendly aliases used by the demos.                                             | <code>appkit.h</code> |
| <code>get_ticks</code>            | Friendly aliases used by the demos.                                             | <code>appkit.h</code> |
| <code>msleep</code>               | Friendly aliases used by the demos.                                             | <code>appkit.h</code> |
| <code>pump_events</code>          | Friendly aliases used by the demos.                                             | <code>appkit.h</code> |

| Name                           | What it does                                                                                                   | Header                |
|--------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>should_exit</code>       | Friendly aliases used by the demos.                                                                            | <code>appkit.h</code> |
| <code>ax_strcat</code>         | Strings.                                                                                                       | <code>appkit.h</code> |
| <code>ax_app_path</code>       | Strings.                                                                                                       | <code>appkit.h</code> |
| <code>ax_streq</code>          | Strings.                                                                                                       | <code>appkit.h</code> |
| <code>ax_strlen</code>         | Strings.                                                                                                       | <code>appkit.h</code> |
| <code>ax_itoa</code>           | Strings.                                                                                                       | <code>appkit.h</code> |
| <code>ax_fmt2</code>           | Strings.                                                                                                       | <code>appkit.h</code> |
| <code>ax_puts</code>           | The console                                                                                                    | <code>appkit.h</code> |
| <code>ax_putln</code>          | The console                                                                                                    | <code>appkit.h</code> |
| <code>app_ini_load_path</code> | load the .ini file at path (replacing the one loaded) → its number of entries, -1 not opened                   | <code>appkit.h</code> |
| <code>app_ini_load</code>      | the same for <the program's own folder>/filename (kapi_app_dir)                                                | <code>appkit.h</code> |
| <code>app_ini_get</code>       | The value of key in section (0 or "" = the keys before any [section]) of the loaded file, else def.            | <code>appkit.h</code> |
| <code>app_ini_get_int</code>   | The value of key in section as a decimal integer (a sign allowed), def when the key is absent or has no digit. | <code>appkit.h</code> |
| <code>app_ini_count</code>     | the number of entries (key=value lines) of the loaded file                                                     | <code>appkit.h</code> |
| <code>app_ini_section</code>   | the section of the entry i (0-based, in the file's order); "" out of range                                     | <code>appkit.h</code> |
| <code>app_ini_key</code>       | the key of the entry i; "" out of range                                                                        | <code>appkit.h</code> |
| <code>app_ini_value</code>     | the value of the entry i; "" out of range                                                                      | <code>appkit.h</code> |
| <code>ax_load_keymap</code>    | The keyboard layout ("FR", "BE"...)                                                                            | <code>appkit.h</code> |
| <code>lx_low</code>            | c in lower case ('A'..'Z' only)                                                                                | <code>appkit.h</code> |
| <code>lx_len</code>            | the length of s (0 for a null pointer)                                                                         | <code>appkit.h</code> |
| <code>lx_cat</code>            | s appended to d at *n                                                                                          | <code>appkit.h</code> |

| Name                       | What it does                                                                                                                  | Header                |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|                            | (advanced), never past cap, NUL-terminated                                                                                    |                       |
| <code>lx_exists</code>     | 1 if the file at path can be opened ( <code>kapi_open</code> ), else 0                                                        | <code>appkit.h</code> |
| <code>lx_entry</code>      | The i-th "ext = program" line of runners.ini (0-based)                                                                        | <code>appkit.h</code> |
| <code>lx_lists_ext</code>  | Does the value of an app.txt's "games" / "opens" (the extensions after each "System:", or all the words of "opens") hold ext? | <code>appkit.h</code> |
| <code>lx_app_for</code>    | The app (SD:/apps/.app/main) whose app.txt opens files of this extension                                                      | <code>appkit.h</code> |
| <code>lx_runner</code>     | The runner of a file, by its extension (runners.ini, else the built-in list)                                                  | <code>appkit.h</code> |
| <code>lx_cmdline</code>    | The runner's command line                                                                                                     | <code>appkit.h</code> |
| <code>lx_open_as</code>    | A program file                                                                                                                | <code>appkit.h</code> |
| <code>lx_open</code>       | A program file                                                                                                                | <code>appkit.h</code> |
| <code>lx_launch_dir</code> | An app bundle by folder path (".../x.app")                                                                                    | <code>appkit.h</code> |
| <code>lx_launch</code>     | An app by name (SD:apps/.app), with arguments (0 / "")                                                                        | <code>appkit.h</code> |

## appkit/appkit.h

appkit.h -- AppKit: what a program includes to talk to the system.

A program calls the functions declared here (`kapi_*`: the names of AppKit's table, kept as they were); they are AppKit's (SD:/lib/appkit.so, user/Kits/appkit), which the kernel binds to every program. A program never talks to the kernel itself. Their bodies are in `appkit_calls.inc`, beside this file. The structures and constants shared with the kernel come from `<kern/kapi_abi.h>`.

(This header was `user/kapi.h` until 2026-10-05. There is no `kapi.h` any more: what talks to the kernel is in this folder only.)

## how a program reaches the kernel: AppKit (2026-10-05)

The `kapi_*` functions declared below (`KAPI_FN`) are AppKit's (SD:/lib/appkit.so, user/Kits/appkit: the ONE interface between the programs and the kernel -- loaded by the kernel and bound to every program with no call of theirs). A program calls them BY NAME, through the import stubs

linked into it (lib/appkit\_stubs.o). THIS HEADER ONLY DECLARES THEM: their bodies -- the calls through the kernel's table -- are in appkit\_calls.inc (beside this file), compiled into AppKit (appkit/appkit.c) and nowhere else. So the kernel's table can be restructured -- entries moved, removed, merged -- by adapting that file and rebuilding AppKit alone: no program changes.

The one exception: with KAPI\_INLINE (the tests of the table itself: el0test, faulttest), or in a PC build (the simulator's stand-in kernel has a table of its own and no AppKit), the bodies are compiled inline into the program -- the include at this header's end.

What stays inline here makes no call to the kernel (the spin locks, kapi\_clock\_us, the sound ring's writer): an app core may use it.

```
#define KAPI_C extern "C"
#define KAPI_C
#define KAPI_FN static inline
#define KAPI_FN KAPI_C
```

The kernel's kapi version (KAPI\_ABI\_VERSION as the running kernel says it): what a program tests before a call that came late (it was the first word of the kernel's table).

```
unsigned kapi_abi_version (void);
```

The address of a slot of the kernel's table (its index in 8-byte words): for the tests of the table itself. Nothing else needs it.

```
unsigned long long kapi_table_slot (unsigned slot);
```

Window creation flags (must match kern/gui/window.h).

```
#define WIN_FLAG_BORDERLESS (1u << 0) // no title bar / border / close box
#define WIN_FLAG_BACKMOST (1u << 1) // pinned to the bottom of the z-order (shell desktop)
#define WIN_FLAG_TOPMOST (1u << 2) // pinned to the top, never active (the menu bar)
#define WIN_FLAG_TRANSPARENT (1u << 3) // magenta (0xFF00FF) client pixels are see-through
#define WIN_FLAG_SYSTEM (1u << 4) // system component: not listed as an open app (panel taskbar)
#define WIN_FLAG_ALPHA (1u << 5) // (v64, borderless) the pixels' top byte is a transparency
```

(0 opaque .. 255 see-through; clicks there go below)

```
#define WIN_FLAG_FIXED (1u << 6) // (v69) not movable, no title buttons, kept centred
```

Event kinds (must match kern/gui/window.h).

```
#define GUI_EVENT_CLICK 1
#define GUI_EVENT_CHECK_CHANGED 2
#define GUI_EVENT_TEXT_CHANGED 3
#define GUI_EVENT_VALUE_CHANGED 4
#define GUI_EVENT_KEY 5 // key pressed; value = char or KEY_* code
#define GUI_EVENT_CANVAS_CLICK 6 // client-area press; value = (buttons<32)|(x<<16)|y
#define GUI_EVENT_CANVAS_MOTION 7 // drag (button held) over the client area; same value
```

buttons: bit0 left, bit1 right Full pointer stream (ABI v22, opt-in via kapi\_set\_pointer\_handler) for app-side widget toolkits (UIKit.h). value packs (wheel<<48)|(changed<<40)|(buttons<<32)|(x<<16)|y, all client-relative; decode with the GUI\_PTR\_\* macros below.

```
#define GUI_EVENT_PTR_MOVE 8 // cursor moved
#define GUI_EVENT_PTR_DOWN 9 // a button went down (changed = which: 1/2/4)
#define GUI_EVENT_PTR_UP 10 // a button went up (changed = which)
#define GUI_EVENT_PTR_ENTER 11 // cursor entered the client area
#define GUI_EVENT_PTR_LEAVE 12 // cursor left the client area
#define GUI_EVENT_PTR_WHEEL 13 // scroll wheel turned (GUI_PTR_WHEEL = signed notch delta)
#define GUI_EVENT_MENU 14 // menu-bar command chosen (value = item id, ABI v39)
#define MENU_QUIT (-1) // kapi_menu_command id: close the active app
```

Drag & drop (ABI v42) -- to the pointer handler (see kapi\_drag\_begin):

```
#define GUI_EVENT_DROP 15 // dropped on us: GUI_PTR_X/Y + GUI_DND_FLAGS; kapi_drag_data
#define GUI_EVENT_DRAG_OVER 16 // a drag hovers us (GUI_DND_FLAGS & DND_F_LEAVE: it left)
#define GUI_EVENT_DRAG_DONE 17 // to the source: GUI_DND_PID (0 = none) + GUI_DND_FLAGS
#define GUI_EVENT_DISPLAY_RESIZE 19 // (v66) the screen's size changed: GUI_DISPLAY_W / _H (value)
#define GUI_DISPLAY_W(v) ((int) (((v) >> 16) & 0xFFFF))
#define GUI_DISPLAY_H(v) ((int) ((v) & 0xFFFF))
#define GUI_EVENT_WINCTL 18 // (v64) a title button: value KAPI_FRAME_MENU (the window menu)
```

or KAPI\_FRAME\_MAXIMISE (also a double click on the title)

```
#define GUI_DND_FLAGS(v) (((unsigned) (((unsigned long long) (v) >> 32) & 0xFF))
#define GUI_DND_PID(v) ((int) (((unsigned long long) (v) & 0xFFFFFFFF))
#define DND_F_COPY 1 // Ctrl held: copy instead of move
#define DND_F_CANCEL 2 // DRAG_DONE: cancelled (Esc)
#define DND_F_DESKTOP 4 // DRAG_DONE: dropped on the desktop / no window
#define DND_F_LEAVE 8 // DRAG_OVER: the drag left this window
#define DND_TEXT 1 // payload types (= CLIP_TEXT / CLIP_FILES)
#define DND_FILES 2
#define MOD_CTRL 1 // kapi_get_modifiers
#define MOD_SHIFT 2
#define MOD_ALT 4
#define WIN_MENU_MAX_USER 2048 // max menu spec length (= the kernel's WIN_MENU_MAX)
#define GUI_PTR_Y(v) ((int) (((unsigned long long) (v) & 0xFFFF))
#define GUI_PTR_X(v) ((int) (((unsigned long long) (v) >> 16) & 0xFFFF))
#define GUI_PTR_BUTTONS(v) ((int) (((unsigned long long) (v) >> 32) & 0xFF)) // held mask
#define GUI_PTR_CHANGED(v) ((int) (((unsigned long long) (v) >> 40) & 0xFF)) // 1 left/2 right/4 mid
#define GUI_PTR_WHEEL(v) ((int) (signed char) (((unsigned long long) (v) >> 48) & 0xFF)) // +fwd/-back
```

Logical key codes (GUI\_EVENT\_KEY value). Printable keys are their ASCII value.

```
#define KEY_BACKSPACE 8
#define KEY_TAB 9
#define KEY_ENTER 13
#define KEY_UP 0x100
#define KEY_DOWN 0x101
#define KEY_LEFT 0x102
```

```

#define KEY_RIGHT      0x103
#define KEY_HOME       0x104
#define KEY_END        0x105
#define KEY_PGUP       0x106
#define KEY_PGDN       0x107
#define KEY_DEL        0x108
#define KEY_F1         0x110 // .. KEY_F12 = 0x11B (KEY_F1 + n - 1)
#define KEY_F12        0x11B

```

## windowing

This process's window (one a process), a client area of w x h pixels titled t, placed by the system → its canvas (0x00RRGGBB pixels), 0 on failure (bigger than the screen, no memory).

```
unsigned * kapi_create_window (int w, int h, const char *t);
```

This process's window at x, y (the frame's top left, negative = placed by the system) with the flags f (WIN\_FLAG\_\*) → its canvas, 0 on failure.

```
unsigned * kapi_create_window_ex (int x, int y, int w, int h, const char *t, unsigned f);
```

This window's size set to w x h, within the canvas it was created with (which stays) → the canvas, 0 no window.

```

unsigned * kapi_resize_window (int w, int h);
void kapi_move_window (int x, int y); // this window's frame moved to x, y (screen coordinates)
int kapi_launch (const char *n); // start the app n (SD:apps/<n>.app/main) as a new process -> 1, 0 failure

```

Toggle the app n → 0 it was running and is asked to close (its window's exit flag), 1 it was started, -1 on error.

```
int kapi_toggle_app (const char *n);
```

The running app n's window (one on the current workspace) to the front → 1, 0 not running / no window.

```
int kapi_raise_app (const char *n);
```

The names of the open apps (a window on the current workspace, not a WIN\_FLAG\_SYSTEM one), one a line, into b (s bytes) → how many.

```
int kapi_list_windows (char *b, unsigned s);
```

Every task, one line each " " (state R / S / B / N, kind a = an app or k = a kernel task), into b (s bytes) → how many.

```

int kapi_list_tasks (char *b, unsigned s);
int kapi_kill (const char *name); // kill the app of that name -> 1, 0 (not running, a kernel task, the caller)

```

ps / kill by PID. list\_procs: lines " <a|k> ". kill\_pid: force 0 = clean close, 1 = hard terminate; 1 ok / 0 no such pid / -1 protected.

```

int kapi_list_procs (char *b, unsigned s);
int kapi_kill_pid (int pid, int force);

```

(v91) A process's tree: its descendants (the processes it spawned, the ones they spawned...). op KAPI\_TREE\_LIST → how many descendants pid has (up to cap of their pids into out); KAPI\_TREE\_KILL: pid and all of them terminated now, the leaves first → how many were; KAPI\_TREE\_KILL\_CHILDREN: its descendants only. -KAPI\_ESRCH no such process, -KAPI\_EPERM a kill that would take the caller, -KAPI\_EINVAL, -KAPI\_ENOSYS (a kernel before v91). A terminal closing a tab: its shell and all it runs.

```
int kapi_proc_tree (int pid, int op, int *out, unsigned cap);
```

Keyboard layout: switch among the compiled-in country maps; read the current one.

```
int kapi_set_keymap (const char *name);
int kapi_get_keymap (char *b, unsigned s);
```

Modal-dialog button sets (used by the user-side ui::MessageBox in uidialog.hpp; the kernel message\_box/file dialogs were removed -- the toolkit draws its own).

```
#define MB_OK      0
#define MB_OKCANCEL 1
#define MB_YESNO   2
#define MB_YESNOCANCEL 3 // uikit: Yes = 1, No = 2, Cancel / Esc = 0
```

Run an ELF at an absolute path with an argv string (fire-and-forget). 1/0.

```
int kapi_exec (const char *path, const char *args);
```

Framebuffer size in pixels (for edge-pinned borderless windows).

```
void kapi_screen_size (int *w, int *h);
int kapi_wallpaper_generate (unsigned base, int pts, unsigned seed);
```

App-drawn wallpaper: get the shared screen-sized buffer, draw into it, then commit.

```
unsigned * kapi_wallpaper_buffer (int *w, int *h);
void kapi_wallpaper_commit (void);
void kapi_present (void);
unsigned kapi_get_ticks (void);
void kapi_msleep (unsigned ms);
void kapi_yield (void);
void kapi_exit (int s);
```

(The kernel-drawn widget wrappers -- kapi\_add\_/kapi\_widget\_ -- were removed: every app now builds its UI with the user-side uikit.hpp toolkit. See uikit.hpp.)

## events

Run what is pending -- the kapi\_post calls, then this window's events through their handlers -- and return (no wait).

```
void kapi_pump_events (void);
void kapi_wait_for_exit (void); // pump the events (sleeping between them) until this
// window is asked to close
int kapi_should_exit (void); // 1 once this window was asked to close (its close
// box, kapi_toggle_app), else 0
```

## app-drawn text + keyboard

One line of text s in the kernel's font at x, y of this window's canvas, colour c (0x00RRGGBB), the background kept.

```
void kapi_draw_text (int x, int y, const char *s, unsigned c);
```

Draw kernel-font text into an arbitrary app-mapped 0x00RRGGBB buffer (e.g. a window- chrome copy from kapi\_get\_chrome). Transparent background; dst must be a user VA.

```
void kapi_draw_text_buf (unsigned *dst, int dw, int dh, int x, int y, const char *s, unsigned c);
```

Window surfaces for a user-side chrome drawer (ABI v28). Returns 1 + fills \*out (content canvas + active/inactive chrome copies + insets + title), or 0 if no window.

```
int kapi_get_chrome (struct kapi_chrome *out);
int kapi_font_width (void);
int kapi_font_height (void);
void kapi_set_key_handler (gui_handler fn);
void kapi_set_click_handler (gui_handler fn);
void kapi_set_pointer_handler (gui_handler fn);
```

Memory snapshot (KB): total RAM, free, app-owned, page size. Any pointer may be 0.

```
int kapi_meminfo (unsigned long *total_kb, unsigned long *free_kb, unsigned long *app_kb, unsigned *page_kb);
```

ABI v33: firmware-detected board RAM + app page-pool (HIGH zone) total/free, the bytes reclaimed above 4GB, and the high-segment count. All KB; any pointer may be 0. (detected = physical board RAM e.g. 8192 MB; apppool = the zone backing app frames via palloc\_high.)

```
int kapi_ram_detail (unsigned long *detected_kb, unsigned long *apppool_kb, unsigned long *apppool_free_kb, unsigned long *above4g_kb, unsigned *nsegments);
```

ABI v34: scroll-wheel speed = lines scrolled per notch, applied system-wide (clamped 1..16). The theme editor sets + persists it (SD:/etc/theme.txt wheelspeed=N).

```
void kapi_set_wheel_speed (int lines_per_notch);
int kapi_get_wheel_speed (void);
```

Per-process heap: move the break by inc bytes (Unix sbrk); returns the previous break or (void\*)-1. The user allocator (umm.h) is built on this; apps rarely call it.

```
void * kapi_sbrk (long inc);
```

## enumeration + clock

The names of the installed apps (the folders SD:apps/.app), one a line, into b (s bytes) → how many.

```
int kapi_list_apps (char *b, unsigned s);
```

The local date and time (any pointer may be 0) → 1 a real date, 0 the clock is not set yet (the time since the boot).

```
int kapi_get_datetime (int *y, int *mo, int *d, int *h, int *mi, int *se);
int kapi_app_dir (char *b, unsigned s); // "SD:apps/<this process's name>.app/" into b (s bytes) -> its length
```

## console + files

b (the first 128 bytes of its n) to the kernel's log as a line of "app" → n, -1 a bad buffer (fd is not used).

```
int kapi_write (int fd, const void *b, unsigned n);
```

Open a file to read it (on the card, the RAM volume or a provider's path, relative to the working directory) → its handle, 0 failure.

```
void * kapi_open (const char *p);  
int kapi_read (void *h, void *b, unsigned n); // up to n bytes from the file's  
position (advanced) -> the bytes read (0: the end), -1 error  
unsigned kapi_fsize (void *h); // the file's size in bytes (0xFFFFFFFF: over 4 GB,  
see kapi_fsize64); 0 for a bad handle  
void kapi_close (void *h); // close a file opened with kapi_open
```

save\_file: the whole file written (created / replaced): the bytes written (>= 0; an empty file: 0 -- test < 0 for a failure, or == n), or -1.

```
int kapi_save_file (const char *p, const void *b, unsigned n);
```

Working directory: chdir (1/0) + getcwd. Relative paths in file ops resolve here.

```
int kapi_chdir (const char *p);  
int kapi_getcwd (char *b, unsigned n);
```

## directory listing

```
void * kapi_opendir (const char *p); // open a folder to list it -> its handle, 0  
failure (not a folder)  
int kapi_readdir (void *d, struct kapi_dirent *e); // the next entry (name, size,  
is_dir) into *e -> 1, 0 at the end / on error  
void kapi_closedir (void *d); // close a folder opened with kapi_opendir
```

mkdir / remove / rename: 0 = success, -1 = failure (FatFs result).

```
int kapi_mkdir (const char *p);  
int kapi_remove (const char *p);  
int kapi_rename (const char *from, const char *to);  
void kapi_cursor_pos (int *x, int *y);
```

## stdio / streams / processes

```
void * kapi_pipe (void); // a new pipe (a FIFO in memory) -> its stream handle, 0  
failure  
void * kapi_file_in (const char *p); // a file as a stream to read -> its stream  
handle, 0 failure
```

A file as a stream to write, created / emptied or (append != 0) written at its end → its stream handle, 0 failure.

```
void * kapi_file_out (const char *p, int append);
```

Up to n bytes from a stream (a pipe waits for its writer) → the bytes read, 0 the end / a bad handle.

```
int kapi_stream_read (void *h, void *b, unsigned n);
```

Up to n bytes from a stream without waiting → > 0 the bytes read, 0 the end / a bad handle, -1 nothing yet.

```
int kapi_stream_read_nb (void *h, void *b, unsigned n);
```

n bytes to a stream (a full pipe waits for its reader) → the bytes written, -1 error.

```
int kapi_stream_write (void *h, const void *b, unsigned n);
void kapi_stream_close (void *h); // close a stream handle (its reference to the
stream dropped)
void kapi_stream_eof (void *h); // tell the stream's readers it has ended (the writer
is done); the handle stays open
```

1 if the process of kapi\_spawn has finished (a bad handle too), 0 if it runs (the handle stays, kapi\_wait closes it).

```
int kapi_proc_done (void *proc);
int kapi_stdin_read (void *b, unsigned n); // up to n bytes from this process's
standard input -> the bytes read, 0 the end / none
```

n bytes to this process's standard output (it has none = to the kernel's log, 128 bytes at most) → the bytes written, -1 error.

```
int kapi_stdout_write (const void *b, unsigned n);
```

Start the program at path with the arguments args, its standard input / output the stream handles in / out (0 = none) → its process handle (kapi\_wait, kapi\_proc\_done), 0 failure.

```
void * kapi_spawn (const char *path, const char *args, void *in, void *out);
int kapi_wait (void *proc); // wait for a spawned process's end -> its exit status
(the handle closed), -1 not a process handle
int kapi_get_args (char *b, unsigned n); // this process's argument string into b
(n bytes) -> its length
```

This task's own stdin/stdout stream handles (for a shell wiring children). 0 = none.

```
void * kapi_stdin (void);
void * kapi_stdout (void);
```

Read the next kernel log event (real-time tee). 1 + fills fields, or 0 if empty.

```
int kapi_klog_read (int *sev, char *src, unsigned sc, char *msg, unsigned mc);
```

Verbose kernel logging: toggle (1/0) at runtime + read the current state.

```
int kapi_set_verbose (int on);
int kapi_get_verbose (void);
```

TCP/IP sockets over WLAN (ABI v21). net\_status: 1 + dotted IP into ip[] if the link is up, else 0. tcp\_connect: host = dotted-quad or DNS name; >=0 handle / <0 error. tcp\_send: blocking, the bytes queued / <0 -- a short count: a 5 s timeout after those (they are sent; send the rest again). tcp\_recv: NON-BLOCKING -- >0 bytes, 0 nothing yet, <0 closed/error. tcp\_close: drop the connection.

```
int kapi_net_status (char *ip, unsigned cap);
int kapi_tcp_connect (const char *host, unsigned port);
int kapi_tcp_send (int sock, const void *buf, unsigned len);
int kapi_tcp_recv (int sock, void *buf, unsigned len);
void kapi_tcp_close (int sock);
```

TCP server side (ABI v37). tcp\_listen: listening handle >=0 / <0 (-6 port in use). tcp\_accept: BLOCKS until a peer connects; connected handle >=0 + peer IP / <0.

```
int kapi_tcp_listen (unsigned port);
int kapi_tcp_accept (int listen_sock, char *ip, unsigned cap);
```

Remote screen (ABI v38). `screen_grab`: composite the screen into `dst` (`w*h 0x00RRGGBB`, `w/h = kapi_screen_size`) → 1 / 0, or 2 = unchanged since the previous grab into the same `dst` (left as is: skip the diff). `inject_pointer/inject_key`: input as if from the USB mouse (buttons bit0 L, bit1 R, bit2 M; wheel = notches) / keyboard (cooked string).

```
int kapi_screen_grab (unsigned *dst, int w, int h);
void kapi_inject_pointer (int x, int y, unsigned buttons, int wheel);
void kapi_inject_key (const char *keys);
```

System menu bar (ABI v39). `set_menu`: declare this app's menus (spec lines "M", "\t", "-") + the `GUI_EVENT_MENU` handler -- apps normally use `UIKit::Menu`. `get_menu / menu_command`: for the menu-bar app (active window's spec+title → change serial, 0 = none; send item id, `MENU_QUIT` closes the active app).

```
int kapi_set_menu (const char *spec, gui_handler h);
unsigned kapi_get_menu (char *buf, unsigned cap, char *title, unsigned tcap);
int kapi_menu_command (int id);
```

Named IPC services (ABI v40): `ipc_register` → become service `name` (1 / 0 taken); `ipc_lookup` → its pid or 0. Messages go through `kapi_mailbox_send / _recv` (<= 512 B).

```
int kapi_ipc_register (const char *name);
int kapi_ipc_lookup (const char *name);
```

System clipboard (ABI v40): see `clipboard.h` for the helpers. Types:

```
#define CLIP_TEXT    1    // plain text
#define CLIP_FILES   2    // file/folder paths, '\n'-separated (copied)
#define CLIP_FILES_CUT 3    // same, cut (the paste moves them)
```

The clipboard replaced by the `n` bytes of `d` (64 KB at most, `n 0` empties it) of the type `CLIP_*` → the bytes kept (0 too for a bad pointer, the clipboard then as it was).

```
int kapi_clipboard_set (int type, const void *d, unsigned n);
```

Up to `cap` bytes of the clipboard into `b`, its type and its serial (which changes at every set) → the content's whole length, 0 empty. `b`, type and serial may each be 0.

```
int kapi_clipboard_get (int *type, void *b, unsigned cap, unsigned *serial);
```

Window opacity 0..255 (ABI v40; fades) and end of session (0 = halt, 1 = restart).

```
void kapi_set_window_alpha (int a);
#define SHUTDOWN_HALT    0
#define SHUTDOWN_RESTART 1
void kapi_shutdown (int mode); // end the session: the card unmounted, then halt
                               (SHUTDOWN_HALT) or restart; does not return
```

Full-screen apps (ABI v41): `fullscreen_begin` → a screen-sized `0x00RRGGBB` back buffer (`w/h` filled); the desktop stops drawing and all input (screen coords) comes to you. Draw, present\_fb to show it; `fullscreen_end` (or exiting) gives the desktop back.

```
unsigned * kapi_fullscreen_begin (int *w, int *h);
void kapi_present_fb (void);
void kapi_fullscreen_end (void);
```

Drag & drop (ABI v42). `drag_begin`: while the left button is held (from a pointer-move handler, after a few pixels of motion), drag (type, data <= 4 KB) with `label` on the cursor → 1 / 0. The target gets `GUI_EVENT_DROP` and reads the payload with `drag_data` (copies <= `cap`, returns the

full length); the source gets GUI\_EVENT\_DRAG\_DONE. get\_modifiers: MOD\_\* held now; inject\_modifiers: set them (vncd).

```
int kapi_drag_begin (int type, const void *data, unsigned len, const char *label);
int kapi_drag_data (int *type, void *buf, unsigned cap);
unsigned kapi_get_modifiers (void);
void kapi_inject_modifiers (unsigned mods);
```

Network tools (ABI v43). net\_ping: one ICMP echo (RTT in us, or -1 down / -3 unresolved / -4 timeout / -5 send failed). net\_resolve: DNS → dotted IP (1/0). net\_info: netstat text.

```
int kapi_net_ping (const char *host, unsigned seq, unsigned timeout_ms, char *ip,
unsigned cap);
int kapi_net_resolve (const char *host, char *ip, unsigned cap);
int kapi_net_info (char *buf, unsigned cap);
```

User-space file-system providers (ABI v44, kern/vfs.h): a provider app serves every path starting with its prefix ("FTP:"); the file kapis on those paths reach it as requests.

```
#define VFS_OP_OPEN 1 // path -> status = fid (>= 0); data = u32 file size
#define VFS_OP_READ 2 // a0 fid, a1 offset, a2 length -> data, status = n
#define VFS_OP_CLOSE 3 // a0 fid
#define VFS_OP_LIST 4 // path -> data = (u32 size LE, u8 is_dir, name '\0')*,
status = count
#define VFS_OP_SAVE 5 // path, payload (vfs_req_data) -> status = bytes written
#define VFS_OP_MKDIR 6 // path -> 0 / -1
#define VFS_OP_REMOVE 7 // path -> 0 / -1
#define VFS_OP_RENAME 8 // path -> path2: 0 / -1
int kapi_vfs_register (const char *prefix); // this process serves the paths starting
with prefix -> 1, 0 (taken by another, no room)
```

The next request for this provider into \*req → 1, 0 none (with blocking != 0, after waiting up to 0.5 s for one).

```
int kapi_vfs_next (struct kapi_vfs_req *req, int blocking);
```

Up to cap bytes of the request id's payload (VFS\_OP\_SAVE's data) from offset into buf → the bytes copied, 0 none.

```
int kapi_vfs_req_data (unsigned id, void *buf, unsigned cap, unsigned offset);
```

Answer the request id with its status and len bytes of data (0 = none), its caller woken → 1, 0 (no such request, bad data).

```
int kapi_vfs_reply (unsigned id, int status, const void *data, unsigned len);
```

Wi-Fi scan (ABI v45): the access points around, strongest first (struct kapi\_wlan\_ap: ssid, bssid, security WLAN\_SEC\_OPEN/WEP/WPA/WPA2, channel, freq MHz, level dBm, connected). Takes ~3 s (blocks the caller); returns how many (0 = none / no Wi-Fi).

```
int kapi_wlan_scan (struct kapi_wlan_ap *out, int max);
```

Sound (ABI v46), on the 3.5 mm jack. First acquire the output (1 ok, 0 another app has it, -1 no audio); only the owner plays, and its release -- or its exit -- silences it.

```
kapi_sound_start (voice 0..15, milli-Hz (440 Hz = 440000),
SOUND_SQUARE/SINE/TRIANGLE/
SAW/NOISE, volume 0..255): the note plays until kapi_sound_stop
(voice)
(-1 = all). Short attack / release ramps: no clicks.
```

`kapi_sound_write` (s16 L/R frames at SOUND\_RATE, n): a PCM stream mixed with the voices;  
returns the frames taken (0 = full, retry later) -- audio players.

```
int kapi_sound_acquire (void);  
void kapi_sound_release (void);
```

(RETIRED 2026-10-05: the voices are AudioKit's -- ak\_fm\_start / ak\_fm\_stop / ak\_fm\_instrument, audiokit/audiokit.h --; these three answer -1)

```
int kapi_sound_start (int voice, unsigned millihz, int wave, int volume);  
int kapi_sound_stop (int voice);  
int kapi_sound_write (const short *frames, unsigned n);  
int kapi_sound_status (unsigned *rate, unsigned *free_frames, unsigned *owner);
```

FM (ABI v47): give a voice a 2-operator FM instrument (struct kapi\_fm\_instrument, see kern/kapi\_abi.h), then kapi\_sound\_start (voice, milliHz, SOUND\_FM, volume) keys it on and kapi\_sound\_stop (voice) keys it off (its release rate fades it out).

```
int kapi_sound_instrument (int voice, const struct kapi_fm_instrument *ins);
```

Held keys (ABI v48), for games (key events only report presses): 1 while key is held and this window has the keyboard. key = KEY\_UP/DOWN/LEFT/RIGHT, KEY\_ENTER, 27 (Esc), ' ', 'a'..'z' (US position of the key on a USB keyboard), '0'..'9'. Returns 0 on an older kernel.

```
int kapi_key_held (int key);  
void kapi_inject_key_held (int key, int down);
```

Run a program under another process name (ABI v49): a runner running an app is named after the app (its window, list\_windows, raise\_app). See launch.h.

```
int kapi_exec_as (const char *path, const char *args, const char *name);
```

USB gamepads (v50): the raw state of pad 0..3 (1 = there); user/Include/gamepad.h maps its buttons.

```
int kapi_pad_state (int index, struct kapi_pad *out);
```

App cores (v51): acquire core 2 or 3, run a function of this app there (no kapi call and no malloc in it: compute, and exchange data through memory), poll its state, release.

```
int kapi_core_acquire (void);  
int kapi_core_run (int core, void (*fn) (void *), void *arg, void *stack_top);  
int kapi_core_state (int core);  
void kapi_core_release (int core);
```

The V3D GPU (v52): gpu\_info brings it up (1 = usable, buf says what / why not); gpu\_draw renders a depth-tested triangle list (NDC positions + RGBA8 colours) into pixels.

```
int kapi_gpu_info (char *buf, unsigned cap);  
int kapi_gpu_draw (const struct kapi_gpu_vertex *v, unsigned n, unsigned clear,  
unsigned *pixels, int w, int h, int stride);
```

The GPU's full pipeline (v53): textures (0xAARRGGBB; handle < 0 = a new one, pixels 0 = free), and gpu\_render: batches (each: its vertices, matrix, texture, blending, depth test, culling) drawn in one frame into f→pixels. See kern/kapi\_abi.h.

```
int kapi_gpu_texture (int handle, const unsigned *pixels, int w, int h, int stride);  
int kapi_gpu_render (const struct kapi_gpu_frame *f, const struct kapi_gpu_vertex3
```

```
*v, unsigned nv,
    const struct kapi_gpu_batch *b, unsigned nb);
```

Full screen straight into the displayed framebuffer (v55; after kapi\_fullscreen\_begin): its pixels (stride in pixels), or 0 (keep drawing into the back buffer + present\_fb).

```
unsigned * kapi_fullscreen_direct (int *w, int *h, int *stride);
```

The windows as objects (v56, the remote desktop rdpd): list (bottom to top), a client rectangle's pixels, to the front, close.

```
int kapi_win_list (struct kapi_win_info *out, int max);
int kapi_win_read (unsigned id, int part, int x, int y, int w, int h, unsigned *dst,
int stride);
int kapi_win_raise (unsigned id);
int kapi_win_close (unsigned id);
```

The read position of an opened file (v57): 0, or -1 (an older kernel, a file not seekable).

```
int kapi_seek (void *h, unsigned long long pos);
```

Writable + executable memory for generated code, a JIT (v58): 0 on an older kernel / full. After writing code: clean the D-cache / invalidate the I-cache over it (\_\_builtin\_\_clear\_cache).

```
void * kapi_code_alloc (unsigned long size);
```

A file's whole size (v59: over 4 GB on exFAT; an older kernel: fsize).

```
unsigned long long kapi_fsize64 (void *h);
```

(v71) vol\_info: a volume's room -- "SD:", "SD1:", "RAM:" (the RAM volume: files in memory until the Pi restarts), a path on it → 0 and \*out (total / free / used bytes, its type, KAPI\_VOL\_RAM), -1 no such volume / an older kernel. (RAM: paths work with every file call above from v71.)

```
int kapi_vol_info (const char *path, struct kapi_vol_info *out);
```

(v93) The volumes: SD:, SD1:..SD3: (the card's partitions), USB1:, USB2:, USB3: (USB sticks and disks whole; USB1P1:, USB1P2:... the partitions of one that has several; USB: is USB1:; mounted when plugged in), RAM:. vol\_list: every volume (struct kapi\_volume: its state KAPI\_VST\_, flags KAPI\_VF\_, sizes, type, label, the files open on it; flags KAPI\_VOLS\_ROOM: the free space too) → how many there are (out: up to max). A USB volume pulled out stays listed as KAPI\_VST\_REMOVED (KAPI\_VF\_UNSAFE if it was mounted) until a device takes its place; its gen changes at each event.

```
int kapi_vol_list (struct kapi_volume *out, int max, unsigned flags);
```

vol\_eject: a USB device ("USB1:", or any of its partitions: all of them) made safe to remove -- its written files synced, the device's cache flushed, unmounted → 0; -KAPI\_EBUSY files are open on it (synced, still mounted; KAPI\_EJECT\_FORCE: ejected anyway, they then fail with -KAPI\_EIO), -KAPI\_EINVAL not a removable volume, -KAPI\_ENOENT not mounted.

```
int kapi_vol_eject (const char *vol, unsigned flags);
```

vol\_mount: a USB volume ejected but still plugged in (or SD1..SD3) mounted again → 0 / -KAPI\_E\*.

```
int kapi_vol_mount (const char *vol);
```

vol\_format: the volume emptied, a new file system made (struct kapi\_format: KAPI\_FMT\_AUTO / FAT / FAT32 / EXFAT, the cluster, the label) and mounted → 0, -KAPI\_EPERM (SD:, never;

SD1..SD3 without KAPI\_FMT\_CARD), -KAPI\_EBUSY (files open: KAPI\_FMT\_FORCE), -KAPI\_ENODEV, -KAPI\_EINVAL (the label: 11 characters, none of "+,.,:;=<=>?[]"), -KAPI\_ENOSPC (too small or too big for it), -KAPI\_EIO. The caller waits while it runs (seconds on a big stick).

```
int kapi_vol_format (const char *vol, const struct kapi_format *fmt);
```

(v73) The event pump's kernel half -- what kapi\_pump\_events does, step by step, for a pump of the app's own (a protected app's table runs its pump that way, kern/el0.h). pop\_event: the window's next event → 1 (\*ev; its handler NOT called), 0 none; event\_mods: what kapi\_get\_modifiers says while a key handler runs (ev→mods), returns the previous value to put back; pop\_post: the next kapi\_post call → 1 (\*p, not run), 0 none; pump\_sleep: kapi\_pump\_wait without the pump (→ how many are pending). An older kernel: 0 / 0xFFFFFFFF / 0 / 0 (no sleep).

```
int kapi_pop_event (struct kapi_event *ev);
unsigned kapi_event_mods (unsigned mods);
int kapi_pop_post (struct kapi_posted *p);
int kapi_pump_sleep (unsigned timeout_ms);
```

(v74) A process's system calls (pid 0: the caller): the total, the rate per second, the 8 table slots most called (user/BinUtils/kapi\_names.h names them) → 0, -1 no such process / older kernel, -2 bad pointer.

```
int kapi_proc_stats (int pid, struct kapi_syscall_stats *out);
```

(v75) The POSIX layer's kernel half (docs/POSIX-PLAN.md; the structures and KAPI\_\* values in kern/kapi\_abi.h). Every call returns ≥ 0 on success, -KAPI\_Exxx (newlib's errno value) on failure, and -KAPI\_ENOSYS on an older kernel or until the call is implemented. libonyxposix (user/Runtime/libc/posix) wraps them as the POSIX functions. v75 WP-MEM: virtual memory (vm\_map = mmap, vm\_protect = mprotect, vm\_advise = madvise...), threads with their stack size and TLS (struct kapi\_thread\_attr).

```
long long kapi_vm_map (unsigned long long addr, unsigned long long len, unsigned
prot, unsigned flags);
int kapi_vm_unmap (unsigned long long addr, unsigned long long len);
int kapi_vm_protect (unsigned long long addr, unsigned long long len, unsigned prot);
int kapi_vm_advise (unsigned long long addr, unsigned long long len, int advice);
int kapi_vm_query (unsigned long long addr, struct kapi_vm_region *out);
int kapi_vm_stats (int pid, struct kapi_vm_stats *out);
int kapi_thread_create_ex (const struct kapi_thread_attr *attr);
int kapi_thread_info (int tid, struct kapi_thread_info *out);
```

v75 WP-FILE/PROC: open files with 64-bit offsets (pread / pwrite), stat, unlink / rename of open files, dir\_read (255-character names), non-blocking pipe writes; spawn\_ex / proc\_wait, argv / environment blocks, getpid, clock\_info (CNTPTCT + UTC sample), sleep\_us. file\_read / file\_write: off -1 = at the handle's offset (advanced; O\_APPEND writes at the end), else a pread / pwrite; a write past the end fills the gap with zeros. A file unlinked or renamed while open stays usable through its handles. proc\_wait (h, KAPI\_WAIT\_NOHANG, &st) on a running child → 0 with st.pid set and st.reason -1 (spawn\_ex returns once the child has its pid). get\_argv / get\_env (0, 0) → the block's size. docs/02 §8 "v75: files and processes".

```
long long kapi_file_open (const char *path, unsigned flags, unsigned mode);
long long kapi_file_read (long long h, void *buf, unsigned long long len, long long
off);
long long kapi_file_write (long long h, const void *buf, unsigned long long len, long
long off);
long long kapi_file_seek (long long h, long long off, int whence);
int kapi_file_truncate (long long h, long long size);
int kapi_file_sync (long long h);
```

```

int kapi_file_stat (long long h, struct kapi_stat *out);
int kapi_file_close (long long h);
int kapi_path_stat (const char *path, struct kapi_stat *out);
int kapi_path_unlink (const char *path, unsigned flags);
int kapi_path_mkdir (const char *path, unsigned mode);
int kapi_path_rename (const char *from, const char *to);
int kapi_path_utime (const char *path, long long mtime);
int kapi_dir_read (void *dir, struct kapi_dirent2 *out);
int kapi_stream_write_nb (void *h, const void *buf, unsigned len);
long long kapi_spawn_ex (const struct kapi_spawn_attr *a);
int kapi_proc_wait (void *proc, unsigned flags, struct kapi_proc_status *out);
int kapi_get_argv (char *buf, unsigned cap);
int kapi_get_env (char *buf, unsigned cap);
int kapi_getpid (int which);
int kapi_clock_info (struct kapi_clock_info *out);
int kapi_sleep_us (unsigned long long us);

```

v75 WP-NET: BSD sockets (IPv4 TCP / UDP: non-blocking connect, accept, MSG\_PEEK...) and poll over sockets, streams and files.

```

int kapi_sock_open (int type, unsigned flags);
int kapi_sock_connect (int s, const struct kapi_sockaddr *to);
int kapi_sock_bind (int s, const struct kapi_sockaddr *addr);
int kapi_sock_listen (int s, int backlog);
int kapi_sock_accept (int s, struct kapi_sockaddr *peer, unsigned flags);
long long kapi_sock_send (int s, const void *buf, unsigned long long len, unsigned flags, const struct kapi_sockaddr *to);
long long kapi_sock_recv (int s, void *buf, unsigned long long len, unsigned flags, struct kapi_sockaddr *from);
int kapi_sock_shutdown (int s, int how);
int kapi_sock_close (int s);
int kapi_sock_getopt (int s, int opt, int *value);
int kapi_sock_setopt (int s, int opt, int value);
int kapi_sock_name (int s, int peer, struct kapi_sockaddr *out);
int kapi_poll (struct kapi_pollfd *fds, unsigned n, int timeout_ms);

```

(v76, WP-IPC: docs/POSIX-PLAN.md §14) Local sockets (sock\_pair: STREAM / SEQPACKET / DGRAM; numbers >= KAPI\_SOCKET\_LOCAL\_BASE, served by every sock\_\* call and poll), messages carrying handles (sock\_sendmsg / sock\_recvmsg: struct kapi\_msghdr, struct kapi\_handle\_xfer), shared memory objects (shm\_create / shm\_open / shm\_unlink / shm\_ctl / shm\_map), handle\_close, spawn\_ex2 (handles given to the child) and get\_handles (the child's side). - KAPI\_ENOSYS on an older kernel.

```

int kapi_sock_pair (int type, unsigned flags, int *sv);
long long kapi_sock_sendmsg (int s, const struct kapi_msghdr *m, unsigned flags);
long long kapi_sock_recvmsg (int s, struct kapi_msghdr *m, unsigned flags);
long long kapi_shm_create (unsigned long long size, unsigned flags);
long long kapi_shm_open (const char *name, unsigned oflags, unsigned mode);
int kapi_shm_unlink (const char *name);
long long kapi_shm_ctl (long long h, int op, unsigned long long arg);
long long kapi_shm_map (long long h, unsigned long long addr, unsigned long long len, unsigned prot, unsigned flags, unsigned long long off);
int kapi_handle_close (long long h);
long long kapi_spawn_ex2 (const struct kapi_spawn_attr *a, const struct kapi_handle_xfer *handles, unsigned n);
int kapi_get_handles (struct kapi_handle_xfer *out, unsigned cap);

```

(v77) Program images (docs/02 section 7): a program is loaded once and shared by its processes; its key is its canonical path (lower case, the volume first). image\_preload: loaded

ahead and kept (returns at once; a run of that path then reads nothing from the card);  
 image\_unload: its pin and its name taken away (freed with its last process); image\_list: the live  
 images (path 0), or the one a run of path would map (1 / 0). -KAPI\_ENOSYS on an older kernel.

```
int kapi_image_preload (const char *path);
int kapi_image_unload (const char *path);
int kapi_image_list (const char *path, struct kapi_image_info *out, unsigned cap);
```

(v79) What the running kernel is: "key value" lines (name, abi, built, rev, machine, model, ram;  
 keys may be added) → the text's length; -KAPI\_ENOSYS (and "") on an older kernel. /bin/uname.

```
int kapi_kernel_info (char *buf, unsigned cap);
```

(v80) The cores: each one's role (KAPI\_CORE\_\*), the microseconds it was busy, an app core's  
 owner → 0; -KAPI\_ENOSYS (out zeroed) on an older kernel. Two reads make a load.

```
int kapi_cpu_stats (struct kapi_cpu_stats *out);
```

(v80) The bytes pid's sockets received and sent, its open sockets (pid 0: every process's) → 0; -  
 KAPI\_ENOSYS (out zeroed) on an older kernel.

```
int kapi_net_stats (int pid, struct kapi_net_stats *out);
```

(v81) The pointer's shape over this window (KAPI\_CURSOR\_\*) → the shape it had; -1 on an older  
 kernel (the arrow stays). uikit: uk\_cursor, from a widget's onMouse.

```
int kapi_set_cursor (int shape);
```

(v82) This window can be resized by its frame (min\_w x min\_h: its smallest client area) → 0; -1 on  
 an older kernel, or for a borderless / fixed window. At the release of a drag the pointer handler  
 gets GUI\_EVENT\_WINRESIZE: GUI\_WINRESIZE\_X / \_Y (the frame's new top left), \_W / \_H (the  
 client area's new size) of its value; the app applies them. uikit: Root::setResizable.

```
#define GUI_EVENT_WINRESIZE 20
#define GUI_WINRESIZE_X(v) ((int) (short) ((unsigned long long) (v) >> 48))
#define GUI_WINRESIZE_Y(v) ((int) (short) ((unsigned long long) (v) >> 32))
#define GUI_WINRESIZE_W(v) ((int) (((unsigned long long) (v) >> 16) & 0xFFFF))
#define GUI_WINRESIZE_H(v) ((int) ((unsigned long long) (v) & 0xFFFF))
int kapi_win_resizable (int on, int min_w, int min_h); // this window resizable by  

its frame (on 0: no longer), min_w x min_h its smallest client area -> 0, -1
```

(v83) A shared library (docs/SHARED-LIBS-PLAN.md): "uikit" is SD:/lib/uikit.so, anything with a '/'  
 or a ':' a path → its export table (unsigned version, size; int (\*init) (const TLibImports \*)); then its  
 entries), mapped in this process until it ends; 0 with *err* = -KAPI\_E (-KAPI\_ENOTSUP: the library is  
 older than min\_version; -KAPI\_ENOSYS on an older kernel). Apps do not call this: the library's bind  
 object does, before main (user/Runtime/lib.h).

```
const void * kapi_lib_open (const char *name, unsigned min_version, int *err);
```

(v84) The sound's output: KAPI\_SND\_OUT\_AUTO / \_JACK / \_USB / \_HDMI (-1: only ask) → what  
 plays now, what is asked and which outputs are there (KAPI\_SND\_OUT\_NOW / \_ASKED / \_HAS of  
 the result); -KAPI\_ENOSYS on an older kernel (the jack only). The Sound applet; kept in  
 SD:/etc/sound.ini.

```
int kapi_sound_output (int out);
```

(v85) the sound's mixer: the programs that play (each has a channel), a channel's volume and  
 mute

```
int kapi_sound_clients (struct kapi_sound_client *out, int max);
int kapi_sound_client_volume (unsigned pid, int volume, int mute);
```

(v73) Is this process protected (ELO, kern/el0.h)? Its table's memcpy is then user code, next to the table, instead of the kernel's.

```
int kapi_is_protected (void);
```

the master volume 0..10 and mute (-1: keep) → volume | 0x100 if muted (older kernel: 10, not muted)

```
int kapi_sound_volume (int volume, int mute);
```

wpa\_supplicant.conf read again + DHCP again, no reboot (-1: none / older kernel)

```
int kapi_wlan_reconnect (void);
```

v61: draws with the app's own QPU shaders (user/Libs/v3d/qpu.h builds them): gpu\_program makes / replaces / frees (p = 0) a program, gpu\_render2 draws batches of them (kapi\_abi.h).

```
int kapi_gpu_program (int handle, const struct kapi_gpu_program *p);
int kapi_gpu_render2 (const struct kapi_gpu_frame *f, const float *v, unsigned nv,
unsigned stride,
const struct kapi_gpu_batch2 *b, unsigned nb, const unsigned
*uni, unsigned nuni);
```

(v62) gpu\_render3: the batches' vertices where they are (kapi\_gpu\_batch3: off, stride), x / y framed by the kernel on the way (view: x' = v0 x + v1 w, y' = v2 y + v3 w; 0: as they are)

```
int kapi_gpu_render3 (const struct kapi_gpu_frame *f, const float *v, unsigned
nfloats,
const struct kapi_gpu_batch3 *b, unsigned nb, const unsigned
*uni, unsigned nuni, const float *view);
```

(v63) gpu\_vbuf: memory the GPU reads too -- gpu\_render3 draws vertices there in place (framed in place: view 0 to draw them again; the clipped triangles into the room past nfloats) → 0 none

```
void * kapi_gpu_vbuf (unsigned bytes);
```

(v70) gpu\_texture\_rect: a rectangle x, y, w x h of a texture's pixels (0xAARRGGBB) replaced, the rest kept → 0, -1 no GPU / older kernel, -2 bad arguments. (Also v70: KAPI\_GPU\_F\_ALPHA, a frame's target keeping its alpha; the GPU compositing service over these: user/Libs/gpucomp/gpucomp.h.)

```
int kapi_gpu_texture_rect (int handle, int x, int y, int w, int h, const unsigned
*pixels, int stride);
```

(v64) the windows of the modernised CDE desktop: minimise one (0: mine; back with win\_raise / raise\_app), my window's place and size and the work area (the screen less the menu bar and the dock), resize my window letting its canvas grow (\*stride: its pixels a row; redraw everything, the frame too) → 0 on an older kernel / no memory.

```
int kapi_win_minimise (unsigned id);
int kapi_win_geometry (struct kapi_win_geom *out);
unsigned * kapi_resize_window2 (int w, int h, int *stride);
```

(v65) the workspaces (virtual desktops): kapi\_desk (set, count) shows desk set (-1 keeps it) and sets how many there are (0 keeps it) → the current desk | the count < 8 | a change counter < 16 (KAPI\_DESK\_CUR / \_COUNT / \_GEN); an older kernel: one desk. kapi\_win\_desk: window id (0: mine) to desk n (-1: every desk; -2: only asked) → its desk, -3 none.

```

#define KAPI_DESK_CUR(i)      ((i) & 0xFF)
#define KAPI_DESK_COUNT(i)   (((i) >> 8) & 0xFF)
#define KAPI_DESK_GEN(i)     (((unsigned) (i) >> 16) & 0x7FFF)
int kapi_desk (int set, int count); // show the desk `set` (-1: keep) and set their
number (0: keep) -> KAPI_DESK_CUR / _COUNT / _GEN of the result
int kapi_win_desk (unsigned id, int n); // the window id (0: mine) moved to the desk
n (-1: every desk; -2: only ask) -> its desk, -3 no such window

```

(v66) screen\_set: the screen's resolution now (640 × 480 .. 2560 × 1600, w even); every window kept on the screen and sent GUI\_EVENT\_DISPLAY\_RESIZE → 0; -1 out of bounds; -2 not now (a full-screen app...); -3 the firmware refused it (old size kept); -4 an older kernel. Not kept across a reboot (SD:/cmdline.txt width= / height= are).

```
int kapi_screen_set (int w, int h);
```

(v67) Threads: more tasks in this process (its memory, window, files, sockets), preempted like it; the process ends with its main thread (tid 1). Timeouts in ms: 0 = only try, KAPI\_WAIT\_FOREVER = none. An older kernel: -3 / nothing (a single thread). kapi\_thread\_create (fn, arg, stack\_size (0: 256 KB), name (0: its tid)) → tid ≥ 2, -1 no memory, -2 too many (32); fn's return value is its exit code. kapi\_thread\_join → 0 (\*code), -1 timeout, -2 no such thread, -3 itself. The GUI belongs to the thread that pumps (the main one): a worker hands its result over with kapi\_post (fn, ctx, value), which the pump (kapi\_pump\_events / kapi\_pump\_wait / kapi\_wait\_for\_exit) runs on its thread. kapi\_pump\_wait (ms) sleeps until an event / a post / the close box, then pumps.

```

int kapi_thread_create (int (*fn) (void *), void *arg, unsigned stack_size, const
char *name);
void kapi_thread_exit (int code);
int kapi_thread_join (int tid, unsigned timeout_ms, int *code);
int kapi_thread_self (void);

```

Synchronisation objects (handles > 0; 256 per process). mutex: recursive, released if its owner thread ends; lock → 0, -1 timeout, -2 bad handle. event: manual reset (stays set until reset) or auto (a wait takes it); wait → 0, -1, -2. barrier: count threads meet; wait → 1 for the last one in, 0 the others. kapi\_sync\_close frees any of them.

```

int kapi_mutex_create (void);
int kapi_mutex_lock (int h, unsigned timeout_ms);
int kapi_mutex_unlock (int h);
int kapi_event_create (int manual_reset, int initial);
int kapi_event_set (int h);
int kapi_event_reset (int h);
int kapi_event_wait (int h, unsigned timeout_ms);
int kapi_barrier_create (unsigned count);
int kapi_barrier_wait (int h);
int kapi_sync_close (int h);
int kapi_post (void (*fn) (void *ctx, long value), void *ctx, long value);
int kapi_pump_wait (unsigned timeout_ms);

```

A user-space lock between this process's threads (the allocators, newlib): an atomic swap, and a yield while another thread holds it -- nothing to create, a zeroed int is free. Not recursive. (Threads all run on core 0; the swap is still an exclusive load / store pair, since the timer may preempt a thread anywhere in its own code.) (The PC builds of the apps -- the desktop simulator, the NetSurf bench -- get the compiler's atomics and no cores.)

The core this code runs on (0: the main one; 2, 3: an app core). The kernel publishes it in TPIDRRO\_ELO (v73, kern/el0.h): every app runs at EL0, where MPIDR\_EL1 may not be read (the app would be killed) -- never read MPIDR\_EL1 in an app.

(on an app core -- kapi\_core\_run's code, which makes no kapi call -- it spins instead)

```
void kapi_lock (volatile int *l);  
void kapi_unlock (volatile int *l);
```

(v68) A futex. kapi\_wait\_word (addr, expected, timeout\_ms) sleeps while \*addr == expected → 0 woken / the value differs, 1 timeout (0 ms: only check), -1 a bad address (not 4-byte aligned, unmapped); may wake spuriously: loop on the condition. kapi\_wake\_word (addr) wakes its sleepers → how many. Works across processes on a shared surface (keyed by the physical address). A word changed by an app core (no kapi there) is noticed at the next 10 ms tick. kapi\_thread\_priority (tid 0 self / 1 main / >= 2, prio 0 normal / 1 real time / -1 ask) → the previous one: a real-time thread (an audio pump) runs first whenever it is ready, as long as it sleeps before its time slice ends. An older kernel: -1 / -2.

```
int kapi_wait_word (volatile unsigned *addr, unsigned expected, unsigned timeout_ms);  
int kapi_wake_word (volatile unsigned *addr);  
int kapi_thread_priority (int tid, int prio);
```

(v68) USB MIDI input: class-compliant devices (keyboards, interfaces), found when plugged in, even later. kapi\_midi\_read (ev, max) takes up to max queued events (struct kapi\_midi\_event: time\_us, cable, status, data1, data2, device, length), oldest first, never waits → how many; one queue for the system (256 events). kapi\_midi\_devices () → attached.

```
int kapi_midi_read (struct kapi_midi_event *ev, int max);  
int kapi_midi_devices (void);
```

(v69) kapi\_screen\_native (&w, &h): the monitor's own resolution, from its EDID → 1, 0 unknown. kapi\_set\_timezone (minutes): the local time's offset from UTC at once (-720 .. 840) → 1 ok. WIN\_FLAG\_FIXED (a window's flags): not movable, no minimise / maximise / close, kept centred.

```
int kapi_screen_native (int *w, int *h);  
int kapi_set_timezone (int minutes);
```

The kernel's microsecond clock (CTimer::GetClockTicks: the ARM counter, same formula), the time base of kapi\_midi\_event.time\_us. No kapi call: an app core may read it too.

```
unsigned kapi_clock_us (void);
```

(v68) Low-latency sound, for the sound owner (kapi\_sound\_acquire). kapi\_sound\_config (chunk frames 64..1024 (0: 1024), chunks ahead 1..4 (0: 4)) → the latency now in frames ((ahead + 1) x chunk: 1024 x 4 ~116 ms by default, 256 x 2 ~17 ms), -1 not the owner / older kernel. Keep the sound\_write stream shallow too (it holds up to 0.5 s): write when kapi\_sound\_status's free frames show little is queued. The defaults come back when the owner releases the output. kapi\_sound\_map () → the mapped PCM ring (struct kapi\_sound\_ring, kern/kapi\_abi.h), mixed until the owner releases the output, or 0. It is plain memory: an app core (kapi\_core\_run) fills it with kapi\_sound\_ring\_write, which makes no kapi call.

```
int kapi_sound_config (int chunk_frames, int ahead);  
struct kapi_sound_ring * kapi_sound_map (void);
```

Frames the ring can take now.

```
unsigned kapi_sound_ring_free (const struct kapi_sound_ring *r);
```

Copy up to n s16 stereo frames into the ring → the frames taken (no kapi call: app cores too).

```
unsigned kapi_sound_ring_write (struct kapi_sound_ring *r, const short *frames,  
unsigned n);
```

Reboot the machine (ABI v25). Does not return. Use to apply settings the kernel only reads at boot -- e.g. after wpaconf rewrites SD:/etc/wpa\_supplicant.conf.

```
void kapi_reboot (void);
```

1 if a USB keyboard is attached & ready, else 0 (ABI v26). The `keyb` tool polls this at boot before applying a layout (it can run before USB enumeration finishes).

```
int kapi_kbd_ready (void);
```

Load a keyboard layout from a .kmap blob (ABI v27): "OKM1" + u16 rows + u16 cols + table. The kernel copies it; the caller frees `data`. 1 on success. See /etc/keymaps.

```
int kapi_set_keymap_data (const char *name, const void *data, unsigned len);
```

Hardware RNG (ABI v30): fill `buf[len]` with random bytes from the Pi's HW RNG. For cryptographic seeding -- the TLS entropy source uses it. Returns bytes written.

```
int kapi_random (void *buf, unsigned len);
```

Shell surfaces (ABI v35): shared 0x00RRGGBB pixel buffers for the activity-shell compositor. The shell creates one sized to a viewport, passes its id to an app; both map it (same physical frames, own VA), the app draws + presents, the shell composites.

```
int kapi_surface_create (int w, int h);
unsigned * kapi_surface_map (int id);
int kapi_surface_size (int id, int *w, int *h);
void kapi_surface_present (int id);
int kapi_surface_destroy (int id);
```

Activity-shell IPC (ABI v35): the kernel routes opaque {from\_pid,type,bytes} messages between per-process mailboxes. A user compositor calls `kapi_register_shell` to become THE shell; apps post to it with `kapi_shell_request`; the shell replies / pushes async events with `kapi_mailbox_send`; both drain with `kapi_mailbox_recv` (fills `*from_pid` / `*type`, returns the payload length, or -1 if empty; blocking != 0 waits).

```
int kapi_register_shell (void);
int kapi_shell_request (int type, const void *in, unsigned len);
int kapi_mailbox_send (int target_pid, int type, const void *in, unsigned len);
int kapi_mailbox_recv (int *from_pid, int *type, void *buf, unsigned cap, int blocking);
```

(v89) The pointer's shape shown now, whatever window it is over (KAPI\_CURSOR\_\*: the arrow, the hand over a link, the I bar over text, the arrows of a frame's edge...) -- for a remote desktop, which shows it on the other machine (rdpd → Onyx Remote). -1: not known (the kernel's own window manager does not say; Elegant, the graphics server, does).

```
int kapi_cursor_shown (void);
```

(v89) The graphics server's own door to the kernel (Elegant, SD:/bin/elegant): the display, the raw input, its wait -- KAPI\_WS\_\* (kern/kapi\_abi.h). Not for programs: a program's windows are the calls above, whoever serves them. → >= 0, or -KAPI\_Exxx (-KAPI\_ENOSYS: a kernel before v89).

```
long kapi_ws_ctl (int op, long a0, long a1, long a2);
```

Memory primitives (ABI v36): the kernel's (Circle's) `memset/memcpy/memmove`. Real symbols (AppKit's; weak ones in a program that has the bodies inline) so the linker can see them: the freestanding app Makefiles alias the C names onto them (-Wl,--defsym,memset=kapi\_memset ...), which resolves the calls GCC emits on its own (array/struct init and copies).

```

void *kapi_memset (void *dst, int c, unsigned long n);
void *kapi_memcpy (void *dst, const void *src, unsigned long n);
void *kapi_memmove (void *dst, const void *src, unsigned long n);
}
#endif
#endif

```

Friendly aliases used by the demos.

```

unsigned *create_window (int w, int h, const char *t);
void      present (void);
unsigned  get_ticks (void);
void      msleep (unsigned ms);
void      pump_events (void);
int       should_exit (void);

```

## AppKit's small services (appkit\_lib.inc; they were user/applib.h until 2026-10-05)

For every program, with or without a C library.

Strings. ax\_strcat: src appended to dst at \*pos (advanced), never past cap, always NUL-terminated. ax\_app\_path: "SD:apps/" (a suffix: ".app/icon.bmp"). ax\_streq: 1 = the same. ax\_itoa: a signed int in decimal → its length. ax\_fmt2: two digits, zero-padded, into d[0], d[1].

```

void ax_strcat (char *dst, int cap, int *pos, const char *src);
void ax_app_path (char *dst, int cap, const char *name, const char *suffix);
int  ax_streq (const char *a, const char *b);
int  ax_strlen (const char *s);
int  ax_itoa (int v, char *b);
void ax_fmt2 (char *d, int v);

```

The console: a string / a line to the standard output.

```

void ax_puts (const char *s);
void ax_putln (const char *s);

```

A minimal .ini reader: [section] headers and key=value lines, ';' or '#' comments, the spaces around trimmed. One file loaded at a time, a program (the store is the program's own); a load replaces the one before. Limits: INI\_MAX entries, INI\_STRLEN - 1 characters a name or a value, INI\_BUFSZ - 1 bytes of file.

```

app_ini_load (filename)      <the program's own folder>/filename -> the number of
entries, -1 absent
app_ini_load_path (path)    any file
app_ini_get (section, key, def), app_ini_get_int (...)  section 0 or "": the keys
before any [section]
app_ini_count (), app_ini_section / app_ini_key / app_ini_value (i)  the entries,
in the file's order

```

```

#define INI_MAX      64
#define INI_STRLEN   64
#define INI_BUFSZ    2048
int app_ini_load_path (const char *path);  // load the .ini file at path (replacing
the one loaded) -> its number of entries, -1 not opened
int app_ini_load (const char *filename);  // the same for <the program's own
folder>/filename (kapi_app_dir)

```

The value of key in section (0 or "" = the keys before any [section]) of the loaded file, else def.

```

const char *app_ini_get (const char *section, const char *key, const char *def);

```

The value of key in section as a decimal integer (a sign allowed), def when the key is absent or has no digit.

```
int app_ini_get_int (const char *section, const char *key, int def);
int app_ini_count (void); // the number of entries (key=value lines) of the loaded file
const char *app_ini_section (int i); // the section of the entry i (0-based, in the file's order); "" out of range
const char *app_ini_key (int i); // the key of the entry i; "" out of range
const char *app_ini_value (int i); // the value of the entry i; "" out of range
```

The keyboard layout ("FR", "BE"...): SD:/etc/keymaps/.kmap (kapi\_set\_keymap\_data), else a map the kernel has (kapi\_set\_keymap). Non-zero = done. The keyboard must be up (kapi\_kbd\_ready).

```
int ax_load_keymap (const char *name);
```

## Starting programs (it was user/launch.h until 2026-10-05)

launch.h -- starting programs from user space. The kernel only loads ELF programs; the other formats have a RUNNER (SD:/etc/runners.ini, "extension = program"): a BASIC program (.bas, .bax) is run by SD:/bin/basic -- or an app that says it opens them: an emulator's app.txt names the games it plays, "games = Game Boy Color: gbc; Game Boy: gb" (and "opens = dol" for files it opens that are not games): installing an emulator's package is enough.

|                        |                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------|
| lx_launch (name, args) | an app: SD:apps/<name>.app/main (an ELF), else its main.<ext>                                             |
| lx_open (path, args)   | run by that extension's runner (named after the app)                                                      |
| lx_runner (path, out)  | a program file: an ELF as it is, else by its runner<br>the runner of a file (by its extension), 0 if none |

C and C++ (run, init, the apps).

```
#define LX_INI "SD:/etc/runners.ini"
char lx_low (char c); // c in lower case ('A'..'Z' only)
int lx_len (const char *s); // the length of s (0 for a null pointer)
void lx_cat (char *d, int cap, int *n, const char *s); // s appended to d at *n (advanced), never past cap, NUL-terminated
int lx_exists (const char *path); // 1 if the file at path can be opened (kapi_open), else 0
```

The i-th "ext = program" line of runners.ini (0-based): 1 found, 0 past the end.

```
int lx_entry (int i, char *ext, int ecap, char *prog, int pcap);
```

Does the value of an app.txt's "games" / "opens" (the extensions after each "System:", or all the words of "opens") hold ext?

```
int lx_lists_ext (const char *v, int len, const char *ext, int games);
```

The app (SD:/apps/.app/main) whose app.txt opens files of this extension: 1 + out, 0 if none.

```
int lx_app_for (const char *ext, char *out, int cap);
```

The runner of a file, by its extension (runners.ini, else the built-in list): 1 + out, 0 if none.

```
int lx_runner (const char *path, char *out, int cap);
```

The runner's command line: "path" args (the path quoted when it has spaces).

```
void lx_cmdline (char *out, int cap, const char *path, const char *args);
```

A program file: by its runner if its extension has one, else as an ELF. name: the process name (0: the kernel's, from the path). 1 = started.

```
int lx_open_as (const char *path, const char *args, const char *name);  
int lx_open (const char *path, const char *args);
```

An app bundle by folder path (".../x.app"): its main (ELF), else main. for a runner.

```
int lx_launch_dir (const char *dir, const char *name, const char *args);
```

An app by name (SD:apps/.app), with arguments (0 / "": none). 1 = started.

```
int lx_launch (const char *name, const char *args);
```

(the tests of the kernel's table, and the PC builds: the bodies inline -- see this header's top)